

Tubulointerstitial Diseases

Mark A. Perazella | Mitchell H. Rosner

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KEY POINTS

- Tubulointerstitial diseases are a relatively common cause of acute kidney injury (AKI) (10%–20% of cases) and are likely to be a significant unrecognized cause of chronic kidney disease (CKD).
- Tubulointerstitial injury can be immune mediated, driven by activation of both innate and adaptive immune responses, or nonimmune, with interstitial fibrosis as the final common pathway. Drug-induced acute tubulointerstitial nephritis (ATIN) is most commonly caused by a T-cell-mediated delayed hypersensitivity reaction.
- Autosomal dominant tubulointerstitial kidney disease (ADTKD), formerly known as familial juvenile hyperuricemic nephropathy or medullary cystic disease, is a rare group of inherited disorders that leads to slowly progressive nonproteinuric CKD. The commonest form, due to mutations in the uromodulin gene, *UMOD*, is associated with hyperuricemia and gout.
- Drugs, particularly antibiotics, proton pump inhibitors, and nonsteroidal antiinflammatory drugs (NSAIDs), are the most common cause of ATIN, with immune checkpoint inhibitors being increasingly recognized. Other causes include autoimmune and systemic disorders, such as systemic lupus erythematosus (SLE), Sjögren syndrome, and sarcoidosis, and infections such as *Legionella*, tuberculosis, polyoma virus, hantavirus, and leptospirosis.
- ATIN presents with minimal symptoms and an acute or subacute rise in serum creatinine. The classic triad of fever, rash, and eosinophilia is uncommon and seen predominantly in drug-induced ATIN due to β -lactams and sulfonamides. Urine sediment findings and urine eosinophils, and imaging tests such as gallium scanning, have limited sensitivity and specificity, so kidney biopsy is generally required to make a definitive diagnosis.
- Early use of steroids is generally recommended for the treatment of drug-induced ATIN if there is no improvement after drug discontinuation, but this is based on limited evidence from small, observational studies. Failure to diagnose and treat ATIN early may result in increased kidney fibrosis and incomplete recovery of kidney function.
- Chronic tubulointerstitial nephritis (CTIN) may develop as a late complication of ATIN, or may be the initial manifestation of exposure to toxins, such as combination analgesics, aristolochic acids, heavy metals, and lithium, or systemic and autoimmune diseases, such as sarcoidosis or Sjögren syndrome.
- CKD of uncertain etiology is a newly recognized cause of CTIN that affects predominantly young, male agricultural workers in hot climates and is a major health concern in certain countries including parts of Central America, Sri Lanka, India, and Egypt.

INTRODUCTION

The renal interstitium consists of the intertubular, extra-glomerular, and extravascular spaces of the kidney. This renal compartment can be affected by a diverse group of diseases that can elicit local inflammation, tubular cell damage, and possibly lead to fibrosis and chronic, progressive kidney disease. Tubulointerstitial diseases are traditionally referred to as interstitial nephritis and can take two broad forms: acute and chronic. Acute forms present with rapid deterioration in renal function along with marked interstitial inflammatory response characterized by mononuclear cell infiltration, interstitial edema, and varying degrees of tubular cell damage. Chronic forms are more indolent and are characterized by interstitial mononuclear cell infiltration, tubulointerstitial fibrosis, and atrophy. There is some overlap between the acute and chronic forms, especially if acute disease is not recognized and treated in a timely manner, which allows for progression to chronic kidney disease (CKD).

PRIMARY VERSUS SECONDARY TUBULOINTERSTITIAL DISEASE

Another classification for tubulointerstitial disease is to consider whether the pathologic process in this compartment is either primary or secondary in etiology. Primary causes include those etiologies of acute and chronic tubulointerstitial injury (such as drug-induced interstitial nephritis, toxin- and heavy metal-induced chronic interstitial disease, and autoimmune-mediated diseases) that directly injure the renal tubules and surrounding interstitium. Injury typically occurs through a variety of pathologic processes, including immune cell, cytokine, and complement-mediated mechanisms, and may be acute or chronic in nature.

However, tubulointerstitial injury is also present with multiple forms of kidney disease, including those that do not primarily involve this compartment (i.e., secondary). Examples of this secondary tubulointerstitial disease are seen in patients with glomerulonephritis, diabetes mellitus, hypertension, atherosclerosis, and cystic renal diseases.^{1,2} More recently, tubulointerstitial changes have been noted as a secondary process after acute kidney injury (AKI).¹⁻⁴ Furthermore, aging itself leads to significant changes and progressive fibrosis in the tubulointerstitial compartment.¹⁻⁶ As a final common pathway, these tubulointerstitial changes correlate with impaired renal function and the presence of renal fibrosis on biopsy.^{1,2}

The importance of these secondary interstitial changes associated with many kidney diseases was first recognized in 1870 when it was hypothesized that interstitial changes documented in Bright disease were responsible for kidney scarring and shrinkage associated with end-stage renal disease (ESRD).⁷ Subsequently, it has been found that in almost all forms of progressive experimental and human CKD, a prominent inflammatory infiltrate exists within the interstitial compartment. The extent of these infiltrates, number of fibroblasts present, and area of fibrosis all correlate with progressive decline in renal function.⁸⁻¹⁰ Several studies have pointed out the prognostic significance of severe injury of

the tubulointerstitium in lupus nephritis, membranous nephropathy, immunoglobulin A (IgA) nephropathy, and other glomerular diseases.¹⁰⁻¹² There is an association between the degree of renal impairment and extent of tubulointerstitial damage in patients with glomerular disease.¹⁰ In 50 cases of persistent glomerulonephritis, the correlation between the extent of tubular lesions and alterations in renal function was far greater than the changes in glomerular structure on renal function.¹⁰ These findings suggested, in chronic glomerulonephritis, that interstitial damage has much more effect on the glomerular filtration rate (GFR) than structural injury in the glomeruli. Subsequent work also highlighted the importance of tubulointerstitial changes on GFR in a wide variety of glomerulopathies.¹³

Several mechanisms may explain how tubulointerstitial disease affects renal function. The simplest explanation is that tubular obstruction from interstitial inflammation and fibrosis impedes urine flow, increases intratubular pressure, and eventually lowers glomerular filtration.^{13,14} A second possible mechanism implicates a reduction in the volume of peritubular capillaries due to progressive fibrosis and, in this setting, the tubulointerstitial compartment becomes relatively avascular and ischemic.¹⁵ As a result of the increase in vascular resistance in the postglomerular segment, hydrostatic pressure in glomerular capillaries also increases, impairing glomerular arteriolar outflow. A third possibility is an alteration in tubuloglomerular feedback leading to GFR reduction.¹⁶ The presence of edema, inflammation, and fibrosis in the renal interstitium, by increasing the interstitial pressure, may lower the sensitivity of the feedback mechanism, possibly through local control of production of vasoactive substances such as angiotensin-II (Ang-II), nitric oxide, and prostaglandins.^{16,17} When tubulointerstitial fibrosis develops, the autoregulation of renal blood flow is also disrupted permanently, leading to falls in GFR.¹⁸ Fourth, glomerular-tubular disconnection, or finding atrophic tubules and glomeruli no longer connected to each other, is a well-recognized consequence of tubulointerstitial injury.¹⁹ This may be particularly true with injury to the proximal tubule.¹⁹⁻²¹ This is consistent with evidence from human studies showing a positive correlation between the fractional volume of the interstitium, percentage of proximal tubule without connection to a glomerulus, and decline of renal function in patients with chronic pyelonephritis.²² Collectively, all these pathophysiologic processes are interrelated and highlight the importance of secondary fibrotic changes in the interstitium as a critical mechanism for progression of CKD.

MECHANISMS OF TUBULOINTERSTITIAL INJURY

Tubulointerstitial injury is caused by a number of processes, many of which share various components that ultimately promote tubulointerstitial changes.^{23,24} Primary innate and adaptive immune injuries in the tubulointerstitium drive inflammation and, with glomerular injury, there are secondary downstream effects from proteinuria leading to tubulointerstitial nephritis. Persistent inflammation leads to destructive progression with tubulointerstitial fibrosis. The inflammatory process may also be driven by genotypic variations among hosts.

COMPONENTS OF THE TUBULOINTERSTITIUM

A large number of intrinsic cells and structural components, as well as infiltrating cells entering this renal compartment, make up the tubulointerstitium. A dynamic interplay exists between intrinsic and extrinsic cells and tubulointerstitial components. The renal tubules and vasculature, along with a number of endogenous and exogenous cells and the interstitium, fall within the compartment known as the tubulointerstitium.²⁵ In essence, the interstitium is composed of the intertubular, extraglomerular, and extravascular spaces of the kidney, which is surrounded by both tubular and vascular basement membranes.²⁵ Fibrillar pericytes reside at the vascular pole between the macula densa and afferent arterioles, presumably participating in renal autoregulatory responses. Vasa recta and lymphatics within the interstitium are involved in solute exchange, which is driven by a hydrostatic pressure gradient. Separation into cortical and medullary interstitia allows countercurrent medullary solute trapping and the development of the corticomedullary osmotic gradient.

Multiple cell types, extracellular matrix, and interstitial fluid are also important components of the interstitium.²⁵ Proximal tubules, which have antigen-presenting capability and secrete various growth factors and cytokines, directly interact with the interstitium through paracrine and juxtacrine mechanisms.²⁶ These cells may also participate in tubulointerstitial inflammation, which is the result of inflammatory and immunologic stimuli they are exposed to on their tubular and basolateral borders. They are influenced by a number of autocoids, which lead to both growth factor (metalloproteinases and associated inhibitors) production and cytokine (transforming growth factor- β [TGF- β]) secretion.²⁷ The resulting cell-matrix interactions determine extracellular matrix production, stabilization, or removal by matrix-producing fibroblast cells.²⁸

Interstitial fibroblasts, which are of mesenchymal origin, are the most abundant cell within the interstitium.²⁹ They are different from other nonfibroblastic cells of renal mesenchymal origin (pericytes, stem cells, myofibroblasts, and vascular smooth muscle cells) and express matrix-promoting and matrix-degrading proteins that determine net matrix production and the amount of interstitial fibrosis within the tubulointerstitium.³⁰

In addition to fibroblasts, macrophages, lymphocytes, lymphatic endothelial cells, mast cells, and dendritic cells inhabit the interstitium. Circulating monocytes that migrate into the kidneys transform into renal interstitial macrophages. Following recruitment of monocytes to the kidney in tubulointerstitial disorders, they subsequently differentiate into inflammatory M1 or reparative M2 phenotypes.³¹ The balance between M1/M2 phenotypic expression and the interaction with T lymphocytes, fibroblasts, and dendritic cells via growth factors, and cytokines determines the extent of renal fibrogenesis and progression of chronic tubulointerstitial disease. Mast cells are sparse within the interstitium, but appear to be both profibrotic and antifibrotic.³² Dendritic cells are antigen-presenting cells derived from bone marrow-derived precursor cells that migrate into the interstitium.³³ These cells interact with T cells within the interstitial lymphatic network to process antigens and elicit inflammatory responses.³⁴ They may also engage antigens presented by

proximal tubular cells, which constitutively express major histocompatibility complex (MHC) class II antigens,³⁵ leading to inflammation and tubulointerstitial nephritis. They may also participate in tubulointerstitial injury following glomerular injury.²⁷

IMMUNE-MEDIATED TUBULOINTERSTITIAL DISEASE

When the immune system targets the renal interstitial compartment as a primary process, target antigens are derived from endogenous and exogenous sources. These include endogenous renal cells, tubular basement membrane (TBM), or other extracellular matrix components, or are exogenous antigens processed and presented by native renal cells or minor populations of dendritic cells within the interstitium.

ANTIGENS WITHIN THE RENAL COMPARTMENT

Proximal TBM antigen is the major target of anti-TBM antibody-mediated interstitial nephritis (TIN) in humans.³⁶ Glycoprotein isoforms of 54 to 58 and 40 to 50 kDa are the TIN antigens recognized by anti-TBM antibodies.³⁶ This glycoprotein has affinity for type IV collagen and laminin and likely serves to stabilize the TBM.³⁷

Drugs and/or drug-hapten complexes can also serve as nephritogenic antigens. These antigens, along with targeting antibodies, can form immune deposits in situ within the interstitium or precipitate as a circulating complex. Examples of this include members of the penicillin family, cephalosporins, and phenytoin. In some cases, antibodies directed against microorganisms may discover cross-reactive epitopes on interstitial components, as in the antibodies to nephritogenic streptococci, which cross-react with type IV collagen. Some anti-DNA antibodies also react with the extracellular matrix components, laminin and heparin sulfate.

ANTIGEN PRESENTATION BY TUBULAR CELLS

T cells recognize foreign antigens when they are digested into small fragments and presented on the surface of antigen-presenting cells (APCs) bound to MHC antigen molecules.³⁸ Recognition of this complex on APCs leads to activation of T cells. CD4⁺ T cells recognize antigens following the processing and presentation of that antigen in conjunction with class II MHC molecules. By contrast, CD8⁺ T cells recognize antigens synthesized by the APC in conjunction with class I MHC molecules. Activation of T cells by APCs expressing processed antigen is also optimized by a number of cell-cell interactions involving cell surface costimulatory molecules and their ligands. Recognition of class II MHC molecules by CD4⁺ T cells generally results in cell proliferation and cytokine expression, whereas CD8⁺ T cells cause target cell death after detecting antigen-class I complex-bearing cells.

Activated tubular epithelial cells release proinflammatory cytokines and CD4⁺ T_H1 chemokines, but not T_H17 chemokines.³⁹ In addition to processing multiple potentially immunogenic peptides, proximal tubular cells could also be exposed to a number of filtered low-molecular-weight proinflammatory cytokines such as interferon- γ (IFN- γ), interleukin-1 (IL-1), and tumor necrosis factor- α (TNF- α) and cytokines secreted by immune cells infiltrating the

interstitium. The ability of renal epithelial cells to present antigen or serve as a target for CD8⁺ T cells depends on the cytokine milieu and whether the net effect of those mediators is proinflammatory or antiinflammatory. Proinflammatory cytokines typically augment the expression of class II MHC molecules on epithelial cells, which is markedly amplified in interstitial nephritis, as is the expression of adhesion molecules such as intercellular adhesion molecule-1 (ICAM-1).⁴⁰

Renal tubular cells also require the augmented expression of costimulatory molecules, which are required for full activation of T cells.⁴¹ The interaction of the costimulatory molecular pair (a T-cell receptor and an APC ligand) results in activation or inhibition of the immune response. Many of the receptors on T cells are members of the Ig superfamily, including CD28, cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), inducible T-cell costimulator (ICOS), and programmed death 1 (PD-1), whereas their receptor ligands are members of the B7 family, which can be induced by proinflammatory cytokines.⁴² Tubular cells can also be induced to express other accessory molecules, such as CD40, ICAM-1, vascular cell adhesion molecule 1 (VCAM-1), and ICOS-L, which are involved in T-cell activation.^{43,44} In addition, renal tubular cell expression of programmed death-ligand 1 (PD-L1), the receptor for PD-1, may have inhibitory effects on T-cell proliferation and/or effector functions.^{45,46} IgA nephropathy, interstitial nephritis, and systemic lupus erythematosus (SLE) nephritis demonstrate significant staining of tubules for B7-H1 (PD-L1),⁴⁷ suggesting that the expression of this costimulatory molecule is upregulated. Thus the number of receptor–ligand pairs that potentially regulate the outcome of renal tubular cell interactions with T cells is large but unpredictable.

CELLULAR INFILTRATES

Cell-mediated immune responses are implicated in the pathogenesis of interstitial nephritis because of both *in vivo* (delayed-type hypersensitivity) and *in vitro* (lymphoblast transformation) evidence of hypersensitivity to specific inciting antigens. However, tubulointerstitial inflammation can result from either antigen-specific stimulation or may develop in the absence of antigenic stimulation.⁴⁸ In the latter case, there is growing evidence for the role of innate immune sensors (TLR2, TLR4, and MyD88 signaling) and the inflammasome complex in initiating tubulointerstitial injury.^{49,50}

The interstitial infiltrate found in most human CKDs consists of a number of different effector cells, including CD4⁺ and CD8⁺ T cells and macrophages, although CD4⁺ T cells predominate.⁵¹ Monocytes and macrophages infiltrating the renal interstitium are also important in the initiation and progression of kidney injury. Macrophages cause cellular damage by generating reactive oxygen species, nitric oxide, complement factors, and proinflammatory cytokines.⁵² These cells can also affect supporting matrix and vasculature through the expression of metalloproteinases and vasoactive peptides. Macrophages may also have a beneficial role in interstitial injury, serving as markers of disease remission⁵³ or having an antifibrotic role.⁵⁴ Clearly, interstitial macrophages have phenotypic and functional heterogeneity in a number of interstitial injury models.

B and T lymphocytes often accompany macrophages within the interstitium. The composition of the interstitial infiltrate is similar, whether the initiating cause of injury is chronic ischemia,⁵⁵ autoimmune tubulointerstitial nephritis,⁵⁶ or

protein overload.⁵⁷ It is possible that immune responses to injured interstitial cell neoantigens are a final common pathway for interstitial injury. In remnant kidney models, infiltration of the interstitium with macrophages and lymphocytes correlates with functional parameters of kidney failure, and improvement occurs with immunosuppressive treatment,⁵⁸ supporting the importance of immune-mediated injury in progressive interstitial disease observed in nonimmune forms of primary injury.

The histologic appearance of human interstitial disease can vary from granulomatous interstitial nephritis with an intense cellular infiltrate to sparse infiltrates. Although this variable appearance may reflect different stages of an immune-mediated lesion or different target antigens, it may also reflect the biologic activity of discrete populations of activated T cells.⁵⁹ Although interstitial injury can be induced by a single T-cell clone, the resultant damage to interstitial and tubular cells is most likely the end result of interactions among many cell types. The cytotoxic activity of renal antigen-reactive T-cell clones may ultimately account for tubular cell destruction and resultant tubule atrophy.

IMMUNOLOGY AND PATHOGENESIS OF DRUG-INDUCED ACUTE TUBULOINTERSTITIAL NEPHRITIS

Drug-induced acute tubulointerstitial nephritis (ATIN) is believed to be primarily a T cell-mediated delayed hypersensitivity reaction, falling into the type IV classification.^{60–65} Renal susceptibility to hypersensitivity is related to the high blood delivery as well as local drug metabolism. Hypersensitivity can be renal limited without systemic manifestations. Four phases characterize this reaction: (1) antigen recognition; (2) antigen presentation; (3) immune regulatory; and (4) effector phases.

Antigen recognition occurs after a drug induces antigenicity by an assortment of mechanisms. Drugs can act as haptens and bind to intrinsic renal proteins, rendering them antigenic.^{60,61} Drugs can also act as prohaptens, which require metabolism by the kidney or liver to become haptens.⁶⁴ Another drug mechanism of antigenicity involves making a native protein immunogenic by damaging tubulointerstitial structures, thereby producing neoantigens.⁶⁶ Various medications are also structurally similar to native kidney antigens (molecular mimicry) and can incite an immune reaction.

As reviewed previously, resident dendritic cells, interstitial macrophages, and tubular cells can function as APCs in the kidneys.^{66–71} Dendritic cells search for foreign material at the basolateral aspects of tubular cells and peritubular capillaries. Although phagocytosis of various native and foreign substances is a major function of macrophages, they also play a role in antigen presentation. Renal tubular cells have inducible MHC class II expression when they come into contact with foreign antigens, facilitating antigen presentation. All of these APCs present antigens to T cells located within the kidneys and regional lymph nodes.

Because immune response in the kidney is self-regulatory, drug-induced ATIN tends to be a rare phenomenon. Self-regulation involves activation of suppressor T cells and downregulation of MHC class II expression.⁶³ Use of checkpoint inhibitors, which activate quiescent T cells to destroy cancer cells, is an example of immune regulation disturbed by medications.

Table 35.1 Drug-Induced Hypersensitivity Reaction Classification

Type IVa
T_H1 cell
IFN- γ /IL-12
• Monocyte/macrophage stimulation
• Synthesis of complement-fixing antibody and complement-mediated injury
• Costimulate proinflammatory responses (tumor necrosis factor, IL-12)
• Costimulate CD8 T-cell responses
Type IVb ~the most common
T_H2 cell
IL-4, IL-13, IL-5
• B-cell production of IgE and IgG4
• Macrophage deactivation
• Mast cell and eosinophil responses
Type IVc
Cytotoxic T cells
• Injury via perforin/granzyme and FasL-dependent processes
Type IVd
T cell (IL-8 and GM-CSF)
• Neutrophil-mediated inflammation/sterile PMN-rich inflammation

FasL, Fas ligand; GM-CSF, granulocyte–monocyte colony stimulating factor; IFN, interferon; Ig, immunoglobulin; IL, interleukin; PMN, polymorphonuclear; T_H , T helper.

The effector phase of the immune system promotes interstitial inflammation and tubular injury via infiltration of lymphocytes, neutrophils, macrophages, mast cells, and eosinophils. Of these, T lymphocytes constitute 80% of the infiltrating cells.^{64,65,71,72} In its most common form, drug-induced ATIN is a type IV(b) hypersensitivity reaction (Table 35.1) with production of IL-5 and presence of systemic and tissue eosinophilia.^{64,71,73} However, different drugs can cause ATIN via different pathways as seen with nonsteroidal antiinflammatory drugs (NSAIDs), which lack interstitial eosinophils implicating a non-type IV(b) drug-hypersensitivity reaction.⁷²

Immune-complex deposition is a rare cause of ATIN with drugs. However, anti-TBM antibody deposition occurs with medications such as methicillin, rifampin, allopurinol, and phenytoin.⁵¹ The deposits generally consist of IgG and complement factor C3 and are deposited along the TBM in a linear or granular fashion. Finally, direct drug interaction with T cells can induce ATIN. Interaction of unprocessed drugs with T cells via T-cell receptor activates T cells, which also play a role in drug hypersensitivity.^{64,74}

NONIMMUNE-MEDIATED TUBULOINTERSTITIAL DISEASE

In addition to immune-mediated tubulointerstitial injury, a variety of nonimmune-mediated processes promote damage to this compartment.

GLOMERULAR-RELATED EVENTS

Glomerular diseases incite tubulointerstitial injury through multiple pathways.⁷⁵ These include (1) impaired glomerular

permselectivity that allows for entry of substances that are morphogenic to tubuli^{75,76}; (2) intraglomerular hypertension damages the nephron,⁷⁷ and alternatively, glomerular hypoperfusion may diminish postglomerular blood flow and cause tubular ischemia; (3) immunologic mechanisms in glomeruli may incur the loss of tolerance and instigate tubulointerstitial injury⁷⁸; and (4) inflammatory mediators enter the interstitium through the juxtaglomerular apparatus.⁷⁹ Subsequent nephron loss caused by injury to glomeruli and attached tubules may facilitate metabolic adaptations in surviving nephrons that induce tubulointerstitial injury through the renin–angiotensin system.⁸⁰

Other possible mechanisms include (1) misdirected filtration and (2) crescentic cell proliferation. According to the misdirection mechanism, extension of a proteinaceous crescent into the outer aspect of the proximal tubule expands into the space between the tubular epithelium and TBM and may spread along the entire proximal convolution within this space. This process may cause tubulointerstitial inflammation and injury.^{81,82} The other mechanism is the encroachment of a growing cellular crescent on the glomerulotubular junction, where the initial segment of the proximal tubule is incorporated into the crescent.^{82,83} This results in nephron loss and subsequent fibrosis, which some consider a reparative process important for the maintenance of renal structure rather than a determinant of further injury.⁸⁴

Ultimately, injury that spreads beyond the confines of the glomerulus allows inflammation to extend within the tubulointerstitial compartment. Chronic injury and the accompanying fibrosis, driven by glomerular processes, provide a pathway for the transmission of more inflammation into areas previously uninvolved by original disease.

PROTEINURIA-INDUCED TUBULAR CELL INJURY

One important mechanism of glomerular–tubulointerstitial interaction is through proteinuria. Although proteinuria is considered as a surrogate marker of the severity of underlying glomerular damage, clinical and experimental data suggest that proteinuria is an independent risk factor and plays an important role in the progression of kidney disease.⁸⁵ Mechanisms underlying nephrotoxic injury induced by excessive proteinuria are multifactorial and involve numerous pathways of cellular damage.

Obstruction of tubular lumens by proteinaceous casts and obliteration of the tubular neck by glomerular tuft adhesions may contribute to tubulointerstitial damage. However, the direct effects of filtered macromolecules on tubular cells are important.⁸⁶ Proteins reaching the tubular urine are largely reabsorbed in the proximal segments of tubular epithelium, where they incite injury. Protein overload activates proximal tubular cells into acquiring a proinflammatory phenotype.⁸⁷ Upregulation of inflammatory and fibrogenic genes and production of related proteins occur with exposure of proximal tubular cells to plasma proteins. These include a number of proinflammatory and profibrogenic cytokines and chemokines.^{88–90} Apoptosis and autophagy are other mechanisms that underlie protein-induced tubular cell injury.^{91,92} Protein overload causes a dose- and duration-dependent induction of apoptosis in cultured proximal tubular cells.⁹³ It is these processes that are thought to underlie proteinuria-induced tubulointerstitial disease. See also Chapter 30 for an in-depth discussion of this topic.

GLOMERULAR FILTERED GROWTH FACTORS AND CYTOKINES

Many growth factors and cytokines, usually in high-molecular-weight precursor forms or bound to specific binding proteins, are present in nephrotic tubular fluid.⁹⁴ Insulin-like growth factor-1 (IGF-1) and hepatocyte growth factor (HGF) both undergo glomerular ultrafiltration in proteinuric states and interact with proximal tubular cells.⁹⁵ In proteinuric glomerular diseases, TGF- β is also present in early proximal tubular fluid at concentrations that are required for biologic responses.⁹⁶ Ultrafiltered IGF-1, HGF, and TGF- β appear to act on tubular cells through their apical signaling receptors, where they induce increased expression of matrix proteins that most likely contribute to interstitial fibrosis.

ACTIVATION OF COMPLEMENT COMPONENTS

Serum- or tubular-derived complement factors are components of urinary proteins that can be harmful to the tubulointerstitium.⁹⁷ Renal tubular epithelial cells appear most susceptible to luminal attack by C5b-9 because of the relative lack of a number of membrane-bound complement regulatory proteins.⁹⁸ Activation of C5b-9 by tubules is associated with tubular injury characterized by severe cytoskeletal alterations, bleb formation, and cytolysis. Increased production of superoxide anion and H₂O₂ and synthesis of proinflammatory cytokines such as IL-6 and TNF- α also contribute to injury.⁹⁹ Accumulation of complement within proximal tubular cells is followed by local recruitment of infiltrating mononuclear cells.¹⁰⁰ Overall, complement activation appears to play a pivotal role as a mediator of progressive tubulointerstitial damage.

TUBULOINTERSTITIAL FIBROSIS

Fibrosis is the final common pathway leading to CKD, irrespective of initiating events. The process of tubulointerstitial fibrosis involves the loss of renal tubules and the accumulation of fibroblasts and matrix proteins, such as collagen (types I–V and VII), fibronectin, and laminin.¹⁰¹ Cells infiltrating the renal interstitium have long been thought to play a role in the initiation and progression of tubulointerstitial fibrosis; however, fibroblasts have been identified as the principal effector mediating tubulointerstitial fibrosis.¹⁰²

EPITHELIAL–MESENCHYMAL TRANSITIONS IN FIBROSIS

Epithelial–mesenchymal transition (EMT) is a robust source of tissue fibroblasts,^{103,104} which play a critical role in tubulointerstitial fibrosis.¹⁰⁵ Epithelial and endothelial cells that line tubules and ducts have plasticity.^{106,107} Midway through development, these cells can transition into fibroblasts by EMT as part of organ growth.¹⁰⁶ Tubular epithelia and endothelia undergoing EMT during persistent injury in the kidney produce new renal fibroblasts, as well as resident and new fibroblasts that proliferate to expand total numbers.^{103,108} Pericytes that have undergone transition are also a source of fibroblasts.¹⁰⁹ Persistent cytokine activity during renal inflammation and disruption of the underlying basement membrane by local proteases initiates the process of EMT.¹¹⁰

A number of proteins, including Wnt proteins, integrin-linked kinases, IGF-1 and IGF-2, epidermal growth factor (EGF), fibroblast growth factor-2 (FGF-2), and TGF- β are

among the archetypal modulators of EMT.¹⁰⁵ However, TGF- β and EGF provide the strongest stimulus for completion of epithelial transitions. These growth factors or their intermediates enhance the transition of epithelial cells into fibroblasts. EMT also operates in human tubular cells^{111,112} and pathologic renal tissues.^{113,114} Importantly, mature epithelia are in a dynamic but not terminal state of differentiation.¹⁰⁷ As such, morphogenic forces that normally maintain epithelial phenotypes are pitted against countervailing forces trying to weaken that stability. Chronic inflammation into the interstitium destabilizes epithelial tissues by favoring fibrogenesis.

CHRONIC HYPOXIA IN FIBROSIS

One of the most important contributors to the development of tubulointerstitial fibrosis is chronic ischemia.^{115–117} Production of AngII and inhibition of nitric oxide production underlie chronic vasoconstriction, which contributes to tissue ischemia and hypoxia¹¹⁸ and stimulates EMT. Loss of peritubular capillaries (rarefaction) in areas of tubulointerstitial fibrosis may develop due to downregulation of vascular endothelial growth factor, and contributes to further ischemia.^{118,119} Moreover, given that the size of the interstitial compartment determines the diffusion distance between peritubular capillaries and tubular cells, interstitial fibrosis likely makes tubular oxygen supply worse. This interstitial reduction of capillary blood flow, leading to starvation of tubules, may underlie tubular atrophy and loss. Under these conditions, the remaining tubules are subject to functional hypermetabolism, with increased oxygen consumption, which further promotes the hypoxic environment and increases amounts of tubulointerstitial fibrosis.

GENETIC CAUSES OF TUBULOINTERSTITIAL DISEASES

There are a rare group of autosomal dominant disorders where disease-causing mutations lead directly to slowly progressive ESRD that is typified by interstitial fibrosis and tubular atrophy along with thickening and lamellation of the TBM. There is no immunologic component for the pathogenesis of these conditions and there is negative immunofluorescence for complement and immunoglobulins.¹²⁰ In the past, this group of diseases has been known by a number of names, such as familial juvenile hyperuricemic nephropathy and medullary cystic disease. However, a recent consensus conference has provided standard terminology for these disorders based on their genetic mutations¹²⁰ (Table 35.2). Thus far, four genes, encoding tubular proteins, have been identified that when mutated lead to autosomal dominant tubulointerstitial kidney disease (ADTKD): uromodulin (*UMOD*), renin (*REN*), hepatocyte nuclear factor 1 β (*HNF1 β*), and mucin-1 (*MUC1*).^{121–124} It is thought that mutations in *UMOD* and *MUC1* are more frequently encountered.¹²⁰

The clinical manifestations as well as the age at which disease manifestations occur are highly variable in these conditions and make diagnosis a challenge. However, in most cases there is a positive family history of kidney disease and this takes the form of autosomal dominant inheritance. Furthermore, the rate of decline in kidney function is variable and the age at which ESRD occurs can range from 20 to 80 years, with most patients requiring renal replacement therapy

Table 35.2 Autosomal Dominant Tubulointerstitial Kidney Disease

KDIGO Terminology	Gene Mutated	Past Terminology	Clinical Characteristics	Laboratory Characteristics	Histological Characteristics
ADTKD-UMOD	<i>UMOD</i>	– Uromodulin kidney disease – Familial juvenile hyperuricemic nephropathy – Medullary cystic disease type 2	– Early onset gout and occasional renal cysts – Rarely presents during childhood	– Hyperuricemia with low-fractional excretion of uric acid – Low urinary excretion of uromodulin	– Intracellular deposits of uromodulin especially in the thick ascending limb of the loop of Henle
ADTKD-MUC1	<i>MUC1</i>	– Mucin-1 kidney disease – Medullary cystic disease type 1	– No characteristic findings – Occasional renal cysts – Presents later in life	None described	Intracellular accumulation of mucin-1 frameshift protein in distal tubules
ADTKD-REN	<i>REN</i>	– Familial juvenile hyperuricemic nephropathy type 2	– Mild hypotension – Anemia during childhood – Frequently presents during childhood	Hyperuricemia, hyperkalemia, and low urinary excretion of uromodulin	Reduced renin staining in cells of the juxtaglomerular apparatus
ADTKD-HNF1 β	<i>HNF1β</i>	– Maturity-onset diabetes mellitus of the young type 5 – Renal cyst and diabetes syndrome	– Maturity-onset diabetes – Few bilateral renal cysts – Pancreatic atrophy – Genital abnormalities – Frequently presents during childhood and even prenatally	Hypomagnesemia, hypokalemia, liver function test abnormalities	Nonspecific

ADTKD, Autosomal dominant tubulointerstitial kidney disease; *HNF1 β* , hepatocyte nuclear factor-1 β ; *KDIGO*, Kidney Disease Improving Global Outcomes; *MUC1*, mucin-1; *REN*, renin; *UMOD*, uromodulin.

between ages 30 and 50. Proteinuria is typically at low levels and the urinary sediment is usually normal.¹²⁵ Although renal cysts can be seen in these conditions, they are generally found in more advanced cases and are no more frequently seen than in other noncystic renal diseases.^{121–125}

There are differentiating characteristics between the various forms of ADTKD (see Table 35.2). ADTKD-UMOD is characterized in some affected individuals by a decreased fractional excretion (FE) of uric acid, hyperuricemia, and gout, which may be manifested in childhood.¹²⁶ ADTKD-REN is characterized by a transient anemia in childhood as well as an increased risk for volume depletion (due to defects in the renin–angiotensin system).¹²⁷ Patients with *HNF1 β* mutations may have multiple extrarenal manifestations, including maturity-onset diabetes mellitus of the young, and some may not manifest tubulointerstitial disease.¹²⁸ ADTKD-MUC1 patients present with progressive tubulointerstitial fibrosis without unique phenotypic characteristics.¹²⁹

ADTKD-UMOD results from mutations in the *UMOD* gene encoding the most abundant protein in the urine, uromodulin (also called Tamm–Horsfall protein).^{120,121} ADTKD-UMOD is the most commonly encountered subtype of ADTKD. Mechanistically, it appears that mutant uromodulin accumulates in the endoplasmic reticulum of the tubular cells in the thick ascending limb of the loop of Henle, leading to cellular injury and defective trafficking of the Na⁺–K⁺–2Cl[–] cotransporter to the luminal surface.^{130,131} These changes lead to impaired sodium conservation, decreased urinary concentrating ability, and resulting mild volume depletion. It is thought that this volume depletion may drive proximal tubular reabsorption of uric acid and lead to the characteristic

finding of hyperuricemia.^{130,131} Progressive CKD is likely related to tubular cell death and fibrosis. ADTKD-UMOD should be suspected in patients with a strong family history of early-onset gout, hyperuricemia, and progressive CKD with bland urinary sediment. Gout in patients with ADTKD-UMOD may occur in the teenage years and in one case series of 205 patients, 65% of patients had gout; however, some families did not display this feature and instead had asymptomatic hyperuricemia in a higher percentage.¹³¹ CKD in these patients follows a variable course with ESRD developing at a median age of 54 years with a range of 25–70 years.^{126,131} The diagnosis of ADTKD-UMOD can be confirmed with specific genetic testing, and renal biopsy does not have a diagnostic role unless specific immunofluorescence microscopy is performed with antibodies to uromodulin, which will demonstrate abnormal deposition in the tubular cell of the thick ascending limb of the loop of Henle.¹³² No specific therapy is available for ADTKD-UMOD and thus therapy should focus on treatment of gout and prevention of progression of CKD. It remains controversial whether allopurinol therapy may slow the progression of kidney disease, especially when initiated early in the course of the disease.^{133,134} However, given the high risk of developing gout, many clinicians recommend allopurinol therapy in patients when hyperuricemia is found. There is no evidence that angiotensin inhibitors slow the progression of CKD in patients with ADTKD-UMOD.

Mutations in the mucin-1 gene (*MUC1*), which encodes a transmembrane protein expressed in the distal tubule, lead to ADTKD-MUC1. This mutation creates a new peptide that accumulates inside renal tubular epithelial cells.¹³⁵ However, the mechanism of progressive CKD is not well understood.

Patients typically present with unexplained progressive CKD with bland urine sediment. Although a renal ultrasound is typically normal in these patients, they may show medullary cysts, which are not sensitive or specific for this condition. In some cases, a history of inherited CKD may be present. The course of CKD is highly variable with ESRD occurring over a wide range of ages (ages 20–70 years).¹³⁶ Genetic testing for ADTKD-MUC1 is not widely available. Care is supportive and follows the general principles for any CKD patient.

Mutations in *HNFIβ* can lead to a variety of phenotypes because the gene is expressed in numerous tissues and regulates other downstream genes.¹³⁷ A whole spectrum of associated phenotypes has been reported, including genital malformations, autism, epilepsy, gout, hypomagnesemia, primary hyperparathyroidism, liver and intestinal abnormalities, and a rare form of kidney cancer.¹³⁸ The mechanisms involved in the development of tubulointerstitial fibrosis are not elucidated. Genetic testing is available for patients suspected of having ADTKD-HNFIβ. Treatment of the renal manifestations of this syndrome is supportive.

ADTKD-REN is due to mutations of the gene encoding preprorenin. Preprorenin undergoes processing to renin, which exerts multiple effects through the prorenin receptor as well as through its protease function to control angiotensin formation.¹³⁹ Mutations in this disease disrupt the translocation of preprorenin into the endoplasmic reticulum of renin-expressing cells that are found in several segments of the renal tubule and juxtaglomerular apparatus.¹²² The net effect is that preprorenin accumulates in these cells, leading to low renin levels and damage of renal tubular cells in affected individuals.¹⁴⁰ Low renin levels are not diagnostic in this condition. Thus some clinical manifestations of ADTKD-REN are due to the low renin state, including low-normal blood pressure, mild elevations of serum potassium, risk for developing volume depletion and prerenal azotemia, and hypoproliferative anemia with low erythropoietin levels that manifests early in life but resolves with adolescence (only to recur once advanced CKD is present).^{140,141} Hyperuricemia and early-onset gout are also present. Progressive CKD begins in childhood and ESRD may be manifest between ages 30 and 60 years. Diagnosis can be confirmed with genetic testing. For some affected individuals with low blood pressure and hyperkalemia, treatment with fludrocortisone and/or a high-salt diet can be helpful. Because of the low renin state in these patients, they are prone to develop volume depletion and a low-sodium diet should be avoided. Furthermore, they are at increased risk to develop AKI when taking NSAIDs.

Kidney transplantation is an effective therapy in all forms of ADTKD because the disorder does not recur in the transplanted kidney. However, family members should be carefully screened for these mutations prior to donating a kidney.

ACUTE TUBULOINTERSTITIAL NEPHRITIS

GENERAL FEATURES OF ACUTE TUBULOINTERSTITIAL NEPHRITIS

ATIN is an entity that is characterized by histologic findings of interstitial inflammatory cellular infiltrate, variable degrees of tubular injury, interstitial edema, and tubulitis. Councilman defined the renal lesion of ATIN as “an acute inflammation

of the kidney characterized by cellular and fluid exudation in the interstitial tissue, accompanied by, but not dependent on, degeneration of the epithelium; the exudation is not purulent in character, and the lesions may be both diffuse and focal.”¹⁴² This form of sterile interstitial nephritis was observed at postmortem in patients dying of scarlet fever and diphtheria. In addition, “allergic” interstitial nephritis was also found in association with bacterial sepsis and other systemic infectious diseases where there was no direct bacterial infection within the kidney parenchyma. As effective antimicrobial agents such as the sulfonamides and β-lactams were used to treat infections, the incidence and prevalence of infection-associated ATIN gradually decreased while pharmacologic agents became the most common etiology.¹⁴³ In subsequent years, interstitial nephritis from autoimmune processes was described and the role of various tubulointerstitial antigens, nephritogenic T cells, and the fibrogenic processes associated with ATIN was demonstrated.¹⁴⁴

INCIDENCE/PREVALENCE

The true incidence and prevalence of ATIN in the overall population are difficult to estimate and most estimates are obtained from biopsy registries.^{145–148} The incidence is highly variable depending on the clinical scenario and the population studied. For example, in patients undergoing kidney biopsy for any indication, the incidence ranges from 1% to 3% (Table 35.3). In fact, most biopsy series support this incidence.¹⁴⁹ An incidence of 0.7% is observed in asymptomatic patients with abnormal urinary findings (hematuria or proteinuria), whereas hospitalized patients with AKI without a clear-cut cause may have an incidence ranging from 10% to 20% (Table 35.3). Of note, the proportion of biopsies showing ATIN appears to be on the rise, although another explanation is an increased rate of detection. The Spanish Registry (1994–2009) noted that the proportion of kidney biopsies with ATIN increased from 3.6% to 10.5% over the duration of the study.^{146–148,150} Similar findings are reported in other studies.¹⁵¹

Medications are now the leading cause of ATIN in the United States, Europe, and other developed countries. Although ATIN can occur in all age groups, it appears to be more common in older individuals. The increase in the number of ATIN diagnoses in the Spanish Registry was driven primarily by older patients (12.3%) versus younger patients (1.6%).¹⁴⁶ A U.S. study reported that the proportion of biopsies showing ATIN in the elderly was 19%, which is higher than seen in the general population.¹⁴⁷ It may be that the increase in biopsy-proven ATIN seen in the elderly patients reflects either polypharmacy with greater drug exposure or other inciting factors in this group.

CLINICAL HISTORY

The clinical presentation of ATIN in patients is quite varied and often challenging to clinicians trying to sort out the correct diagnosis. The presentation can be hospital-acquired AKI, a subacute decline in kidney function in hospitalized or outpatients, or only abnormal urinary findings such as hematuria, pyuria, or proteinuria.^{152,153} The classically taught clinical manifestations of fever, skin rash, arthralgia, and eosinophilia are rarely seen (Table 35.4). The “triad” of ATIN (fever, rash, and eosinophilia) occurs in only approximately 5%–10% of unselected patients and should not be relied on

Table 35.3 Kidney Biopsy Databases Reporting Acute Interstitial Nephritis From Various Countries

Author, Country	Study Period	Total Biopsies Performed	Overall Incidence of AIN	Incidence of AIN in AKI
Haas et al., USA ¹⁴⁷	1991–1998	259	N/R	18.6%
Gesualdo et al., Italy ¹⁴⁸	1996–2000	14,607	5.3%	NR
Davison and Jones, UK ¹⁴⁹	1978–1998	7161	2.2%	6.5%
Muriithi et al., USA ¹⁵⁰	1993–2011	7575	1.8%	NR
Clarkson et al., USA ¹⁵¹	1988–2001	2598	2.6%	10.3%
Valluri et al., Scotland ²⁵²	2000–2012	3604	4.7%	NR
Prendecki et al., UK ²⁵³	1998–2011	3983	5.9%	NR

A/N, Acute interstitial nephritis; AKI, acute kidney injury; NR, not reported.

Table 35.4 Clinical and Laboratory Manifestations of Acute Interstitial Nephritis

Manifestation	Comments
Clinical Features	
Manifestations of allergic reaction: fever, rash, and eosinophilia	Triad is seen in <5%–10%
History of exposure to a potential culprit drug	
Evidence of underlying systemic disease: lungs and skin in sarcoidosis, uveitis in TINU, skin/joint manifestations in SLE, dry eyes in Sjögren syndrome	
Age of onset	TINU and SLE more common in the young, drug-induced more common in elderly
Laboratory Tests	
Subacute or acute rise in glomerular filtration markers (serum creatinine and blood urea nitrogen)	
Elevated inflammatory markers (ESR, CRP)	Not specific for ATIN
Eosinophilia	Inconsistent finding, more common in drug-induced AIN
Elevated IgE level	
Sterile pyuria, white blood cell casts, renal tubular epithelial cells, and granular casts	Limited sensitivity: urine sediment is normal in approximately 20% of biopsy-proven ATIN
Hematuria, rare red blood cell cast	RBC casts are rare
Low-grade (tubular) proteinuria (<1 g/day)	
Urine eosinophils	Sensitivity 20%–30%, specificity 70%–90%, PPV 15%–30%, NPV approximately 85%
Fanconi syndrome (glycosuria, phosphaturia, bicarbonaturia, and aminoaciduria)	Not specific or sensitive for ATIN
Distal tubulopathy (distal RTA, nephrogenic diabetes insipidus, sodium wasting)	Not specific or sensitive for ATIN
Imaging Tests	
Gallium-67 scintigraphy shows enhanced renal uptake	Limited utility due to poor sensitivity and specificity
FDG-PET scanning shows enhanced renal uptake	A few positive cases reported, but studies are lacking
Renal ultrasound and CT scan, may show increased kidney size and echogenicity	Findings nonspecific but helpful to rule out other causes of kidney disease
Kidney Biopsy	
Renal histology reveals inflammatory interstitial infiltrate with tubulitis (and sometimes granuloma)	Gold standard for making the diagnosis

A/N, Acute interstitial nephritis; ATIN, acute tubulointerstitial nephritis; CRP, C-reactive protein; CT, computed tomography; ESR, erythrocyte sedimentation rate; FDG-PET, 2-[¹⁸F] fluoro-2-deoxy-D-glucose-positron emission tomography; NPV, negative predictive value; PPV, positive predictive value; RBC, red blood cell; RTA, renal tubular acidosis; SLE, systemic lupus erythematosus; TINU, tubulointerstitial nephritis and uveitis.

to influence the final diagnosis of ATIN.^{152,153} In fact, this allergic presentation is more often seen in association with certain drugs, such as penicillin derivatives and sulfonamides (compared with NSAIDs), but is by no means a common finding. More commonly, nonspecific symptoms are noted with ATIN.^{154,155} These include generalized malaise, fatigue, weakness, anorexia, and nausea. At times patients will describe myalgias and arthralgias, flank pain, and “feeling feverish.” A pruritic skin rash may develop, raising suspicion for an allergic or drug-related process. However, these are not specific to ATIN and may be seen in many hospitalized patients with or without AKI.^{152–155}

It is important to determine the presence and timing of suspected medication exposure in the clinical evaluation of the patient where ATIN is considered part of the differential diagnosis. As will subsequently be discussed, although any drug may cause ATIN, classic and common agents that should be sought are certain antimicrobial agents such as the β -lactams, sulfonamides, quinolones, and antiviral agents; antiulcer agents such as the proton pump inhibitors (PPIs) and H_2 antagonists; NSAIDs; anticonvulsants; and allopurinol.^{154,155} Table 35.4 provides a list of selected medications that are more commonly associated with ATIN.

PHYSICAL EXAMINATION

Physical examination sometimes points toward a diagnosis of ATIN. Low-grade or spiking fevers that occur in the absence of documented infection should raise suspicion for this diagnosis. In reality, however, it is often quite difficult to sort out such fevers in hospitalized patients who are receiving antibiotics for infection. This includes those with invasive devices in place, such as peripheral or central vein catheters, and indwelling bladder catheters. In addition, fever is not uniformly present, although it occurs more often with ATIN from methicillin and other penicillin derivatives.^{145,146,151–163}

A classic drug eruption, typically morbilliform and involving the trunk, can be very helpful in suggesting the possibility of drug-related ATIN. It is not a sensitive finding, however, and is frequently not present even in the setting of very severe biopsy-proven ATIN. In general, a drug-related rash is reported in approximately 15% to 50% of ATIN cases. This rash is more likely to develop following exposure with drugs that cause a hypersensitivity reaction such as the β -lactams, sulfonamides, and phenytoin. By contrast, rash is rarely seen or completely absent with drugs such as NSAIDs and PPIs.^{145,158,164} Other physical examination findings described with ATIN include palpably enlarged, tender kidneys.^{154,155} However, this is a rarity on examinations. In the absence of an underlying systemic illness, culprit drug exposure, and classic drug eruption, it is extremely difficult to diagnose ATIN in the setting of hospital-acquired AKI in the absence of other supportive data.

In patients with systemic diseases that have been associated with ATIN such as sarcoidosis, SLE, Sjögren syndrome, and other such entities, the presence of active end-organ involvement should raise suspicion of ATIN when renal abnormalities develop.^{153,158,159} In patients with active systemic disease, clues for associated ATIN include various electrolyte acid–base disorders indicative of a tubulopathy (proximal/distal renal tubular acidosis [RTA], hyperkalemia, concentrating defects), an abnormal urinalysis or active urine sediment (hematuria, pyuria, proteinuria, white blood cell [WBC] casts), or the

development of acute, chronic, or acute on CKD.^{152,153,163,164} Table 35.4 notes some of the clinical clues to ATIN in the setting of a systemic disease.

LABORATORY TESTING

In addition to the information provided by clinical history and physical examination in evaluating for possible ATIN, a number of diagnostic modalities are used. A definitive diagnosis of ATIN, especially when drug induced, remains a challenge as the available noninvasive tests lack the required accuracy.^{152–154} As a result, kidney biopsy remains the gold-standard test for diagnosis. Laboratory and imaging tests employed to diagnose ATIN are listed in Table 35.4.

Serum Creatinine Concentration

Increases in serum creatinine concentration, a marker of reduced GFR, can be rapid and indicative of severe AKI. However, the rate of rise in serum creatinine may be slow and not meet the Kidney Disease Improving Global Outcomes (KDIGO) diagnostic criteria of AKI. It is notable that up to 50% patients with kidney biopsy-proven ATIN do not meet currently accepted definitions of AKI.¹⁴⁹ In fact, only 57% of 107 patients diagnosed with ATIN met AKI criteria. Interestingly, >90% of these patients actually met the umbrella term of KDIGO Acute Kidney Disease (AKD) criteria, which is defined as the following: (1) KDIGO AKI criteria, (2) GFR < 60 mL/min/1.73 m² for <3 months, (3) decrease in GFR by $\geq 35\%$ or increase in serum creatinine by >50% for <3 months, or (4) evidence of kidney damage for <3 months.¹⁶⁵ Thus clinicians must remember that an indolent rise in serum creatinine concentration does occur with ATIN and failure to recall this will lead to a delay in recognition of ATIN, which may ultimately be associated with the development of interstitial fibrosis and CKD.

Serum Eosinophil Count

An elevated serum eosinophil count is one of the most helpful serum tests for raising suspicion of ATIN, in particular when due to an allergic or hypersensitivity reaction. Significant eosinophilia may tip the clinician off to an allergic drug reaction and may be very helpful diagnostically for patients with hospital-acquired AKI.^{152–155} Despite the fact that eosinophilia also occurs in other settings, such as cancer, cholesterol emboli syndrome, and vasculitis, which may be associated with AKI, these disease processes are often clinically recognizable.^{152–155} However, as with other tests utilized to evaluate for ATIN, serum eosinophils are not a sensitive or specific finding. Eosinophilia may be only modest and barely recognizable as abnormal or markedly elevated with severe allergic reactions and certain malignancies having eosinophils making up 50%–75% of the total WBC count.¹⁶⁶ An example is the DRESS (drug reaction with eosinophilia and systemic symptoms) syndrome and eosinophilic leukemias. As is observed with drug-related fever and associated morbilliform rash, a significant increase in serum eosinophils in the setting of ATIN is wide ranging (0%–50%), is more common with certain classes of drugs (β -lactams and sulfonamides), and may be absent despite the presence of eosinophil-dominant infiltrate on kidney biopsy.^{145,161,164} As noted previously, the lack of diagnostic utility of the combination of fever, rash, and eosinophilia for ATIN is very disappointing as the triad is seen in only 5%–10% of patients.^{151,161}

Other Serum Tests

Markers of inflammation, such as erythrocyte sedimentation rate and C-reactive protein, are often elevated but are quite nonspecific and not helpful for diagnosis.^{151–153,164} Increased IgE levels may be seen but are an inconsistent finding. Anemia is commonly seen in the setting of ATIN; however, this abnormality is nonspecific and widely prevalent in numerous hospitalized patients, especially those with isolated AKI or AKI superimposed on CKD.^{151–153} Anemia likely results from a number of processes, including loss of erythropoietin production from kidney injury, as well as hyporesponsiveness or resistance to erythropoietin from concurrent inflammation and/or infection.^{151–153} Liver function tests (LFTs) may also be abnormal in patients suffering from ATIN, primarily due to an associated medication-induced hepatitis. However, this finding is exceedingly rare in ATIN, and multiple other disease processes can elevate LFTs in hospitalized patients. Thus the vast majority of these tests are not otherwise useful for diagnosis.

Serum Markers of Tubular Dysfunction

It is possible that tubular injury and abnormalities in serum markers of tubular dysfunction may precede the rise in serum creatinine concentration and development of AKI. Early serum findings of renal tubular dysfunction include hypokalemia/hyperkalemia or RTA. The alert clinician may notice the development of serum chemistry abnormalities that are suggestive of tubular injury and dysfunction. As an example, hyperkalemia with a hyperchloremic metabolic acidosis, which appears out of proportion to the severity of kidney failure, raises the specter of an associated tubulointerstitial injury.^{152,153,164} In addition to these serum abnormalities, other patterns of tubulointerstitial injury such as a Fanconi syndrome, salt-wasting nephropathy, distal RTA, and urinary concentrating defects may be observed with ATIN.^{152,153,164} A classic example of distal RTA associated with ATIN is Sjögren syndrome.

Urinalysis

Urinalysis is considered an important diagnostic test for AKI and is commonly used to understand the underlying cause of AKI in hospitalized patients. This test can provide helpful clues suggestive of ATIN for the clinician.^{152–155,167,168} Because of the nature of tubulointerstitial kidney injury, some urinary abnormality is often present. In general, some degree of low-grade proteinuria is almost universally (>90%) present in most forms of ATIN and generally reflects tubular proteinuria.^{150,169} Trace, 1, or 2+ proteinuria may be seen on the dipstick, unless there is concomitant glomerular injury, such as minimal-change disease or membranous nephropathy associated with NSAIDs, or an underlying glomerulopathy upon which ATIN develops (diabetic nephropathy, and so on).^{152,153} Quantifying the amount of proteinuria with a spot urine protein-to-creatinine ratio generally shows levels <1 g of protein/day, consistent with “tubular” proteinuria.^{145,152,153}

Hematuria is seen in approximately 50% of cases, but is more common, up to 90%, with drugs such as methicillin and the β -lactam class.^{152,153,164,167,168} Pyuria may be seen; sterile pyuria should always raise the specter of ATIN in the appropriate clinical setting. Urinary leukocytes are considered a fairly common urinary abnormality in the setting of ATIN. Early publications on methicillin-associated ATIN noted that

leukocytes were present in up to 90% of cases.^{161,166} In fact, one study noted that approximately 80% of patients with drug-induced ATIN had dipstick pyuria.¹⁷⁰ By contrast, in other forms of ATIN (including drug-induced forms), leukocytes are noted in $\leq 50\%$ of cases.^{159,162} Urinary findings described in 21 cases of biopsy-proven drug-induced ATIN noted red blood cells (RBCs) in 43% and WBCs in 57% of patients, respectively.¹⁶⁷ Thus these studies confirm that although hematuria and leukocyturia are relatively common, clinicians should not erroneously exclude ATIN as a cause of AKI when either hematuria or pyuria is absent.

Urine Eosinophil Test

One of the most common diagnostic tests used for ATIN is the urine eosinophil test. In fact, most clinicians practice with the belief that eosinophiluria is part and parcel of drug-induced ATIN. This clinical approach is based on an early description of nine cases of methicillin-associated ATIN that tested positive for eosinophils, whereas none of 43 patients with AKI from another diagnosis had eosinophiluria.¹⁷¹ In a subsequent small study, eosinophiluria was observed in six of nine patients with drug-induced ATIN.¹⁷² These two small studies form the basis of the widespread use of urine eosinophils for evaluation of ATIN.

Following these two small, positive studies, variable sensitivities and specificities of this test in evaluating for ATIN were noted, raising questions as to the utility of eosinophiluria. As poor visualization of urinary eosinophils was considered a major problem, the Hansel stain rather than Wright stain was pursued to improve test sensitivity.¹⁷³ This was based on the excellent accuracy of this stain in visualizing eosinophils in nasal, bronchial, and ocular secretions of patients with allergic diseases. A small study using this stain noted an improvement in sensitivity to 91%; however, further studies of the Hansel stain^{174,175} demonstrated various ranges of sensitivity and specificity.³⁵ The crux of the problem with this test is that many disorders other than ATIN are associated with significant eosinophiluria. These include cystitis, prostatitis, pyelonephritis, atheroembolic disease, ATN, rapidly progressive glomerulonephritis, allergic granulomatosis, bladder tumors, ileal conduits, and asthma.¹⁷⁵ Many of these also present with AKI.

Despite the unclear utility of the urine eosinophil test, clinicians commonly order this test to evaluate for ATIN in patients with AKI, in particular hospital-acquired AKI. Erroneous decisions based on potentially incorrect results may result—treating with steroids when ATIN is not present or allowing a patient with drug-induced ATIN to continue on the culprit medication. This approach stems from results generated by small studies with many flaws, in particular, the lack of a gold standard for ATIN diagnosis. One of the largest and best studies of the urine eosinophil test was conducted over an 18-year period at the Mayo Clinic on 566 patients who had both urinary eosinophil testing and kidney biopsies performed within the same week.¹⁷⁰ Approximately two-thirds of the biopsy-confirmed ATIN cases were negative for urinary eosinophils. When urinary eosinophils $\geq 1\%$ was used as a positive test, this assay identified only approximately 31% of ATIN cases with a similar sensitivity for ATN (29.0%). The specificity of urinary eosinophils (>1%) for ATIN was 68.2%. A 5% urinary eosinophil cutoff improved specificity (91.2%) but with a concomitant decreased sensitivity (19.8%). Thus

urinary eosinophils should no longer be considered a useful biomarker for ATIN and we recommend abandoning this diagnostic test.^{170,176}

Urine Microscopy and Sediment Examination

Thorough examination of the spun urine sediment performed by an experienced nephrologist is considered to be fairly accurate and tantamount to a “liquid biopsy” of the kidney. In the absence of pyelonephritis, urinary leukocytes and RBCs, along with WBC casts observed in the urine of AKI patients, are highly suggestive of ATIN.^{145,152,153} Unfortunately, WBC casts are not necessarily specific for ATIN as they may be rarely seen with acute glomerulonephritis and acute papillary necrosis.^{145,152,153,155} Other urinary sediment findings also seen with ATIN include renal tubular epithelial cells and casts and granular casts. Tubular epithelial cells and casts reflect tubular cell injury that develops from tubulitis and invading inflammatory cells. In addition, numerous hyaline and granular casts were observed in 18 of 21 patients with biopsy-proven drug-induced ATIN.¹⁶⁷ A surprising finding in this study was the presence of RBC casts in 26% of patients and WBC casts in only 14% of the cases. Furthermore, approximately 20% of patients with ATIN can manifest a normal urinary sediment.^{145,152,153} Thus clinicians should not mistakenly exclude ATIN as a cause of AKI in the absence of pyuria or WBC casts.

Urine Chemistries

Urine chemistries are commonly used in the evaluation of AKI. Urine concentrations of sodium (Na) and urea examined alone or as FEs of Na (FENa) or urea (FEurea) are widely used to assess AKI patients.¹⁶⁸ With some notable exceptions, urine chemistries’ greatest utility is in distinguishing prerenal azotemia from ATN, whereas they are unhelpful for ATIN. Patients with ATIN may have FENa values that are either above or below 1%.^{154,155,168} Although FEurea has not been widely examined in this setting, there is no reason to believe it offers any advantage over FENa.¹⁶⁸ Thus urine chemistries have no role in the evaluation of ATIN.

Novel Urinary Biomarkers

In the field of AKI the role of novel biomarkers in the early diagnosis and prognosis of AKI has been intensely studied. Novel markers of tubular dysfunction, such as monocyte chemoattractant protein-1 (MCP-1), neutrophil gelatinase-associated lipocalin (NGAL), α 1-microglobulin (α 1-MG), and N-acetyl- β -D-glucosaminidase (NAG), have been shown to be elevated in ATIN.¹⁷⁷ However, they have only been studied in drug-induced ATIN patients in comparison with healthy controls,¹⁷⁷ so their value in discriminating between ATIN and other causes of AKI is unknown. Although novel and traditional tubulointerstitial biomarkers, such as urine microalbumin, α 1-MG, matrix metalloproteinase (MMP)-2, MMP-9, urine NGAL, retinol-binding protein, and NAG, correlate with renal prognosis in chronic tubulointerstitial nephritis (CTIN),^{178–180} they have not been adequately studied in ATIN.

Lymphocyte Transformation Test

The lymphocyte transformation test is largely used to evaluate and detect culprit allergens (from several potential candidates) in nonrenal drug-hypersensitivity reactions.¹⁸¹ The test detects

in vitro activation of T cells and measures their proliferation when mixed with nonlethal doses of drug. This test is based on the hypothesis that drugs can directly interact with T-cell receptors in the absence of previous metabolism or need for protein binding. Based on this hypothesis, this test may possibly detect the exact etiology of drug-induced ATIN. Its potential advantages include high specificity (85%–100%) and the ability to simultaneously test multiple drugs. However, the lymphocyte transformation test is technically challenging and highly operator dependent, is dependent on availability of drugs in their pure form, and cannot be performed in the acute stage of hypersensitivity.¹⁸¹ Furthermore, drugs such as NSAIDs that do not normally go through the usual T-cell activation pathway will give false-negative results. As a result, the test is not routinely used in the clinical evaluation of suspected drug-induced ATIN.

Imaging Tests

Ultrasonography and CT Scanning. Renal ultrasound and/or computed tomography (CT) scan imaging is commonly used in the setting of AKI. Kidney imaging with either ultrasonography or CT scan provides structural information such as kidney size and number, cortical echogenicity, and presence or absence of hydronephrosis, cysts, masses, or stones.^{182–184} Thus the utility of these modalities lies with their exclusion of other causes of AKI. Enlarged, swollen kidneys with increased echogenicity on ultrasound are often seen with ATIN. However, this finding is not specific and can be seen with infiltrative diseases, acute tubular necrosis, acute glomerulonephritis, and other etiologies of AKI.^{154,155,183,184} Renal volume was reported to be increased by up to 100% in the setting of ATIN,¹⁸⁵ presumably related to inflammatory cell infiltration and interstitial edema. Similarly, CT scan may show renomegaly in the setting of ATIN, but this test has the same limitations as renal ultrasound.^{182–184} Overall, these imaging tests are neither sensitive nor specific for ATIN, adding little except excluding urinary tract obstruction or another structural form of kidney injury.

⁶⁷Gallium Scanning. Renal imaging using ⁶⁷gallium scanning has been used in the evaluation of ATIN for >30 years.^{186,187} This test is based on the fact that kidneys with ATIN enhance with tracer because ⁶⁷gallium binds to lactoferrin, which is produced and released by and on the surface of leukocytes (primarily lymphocytes) infiltrating the renal interstitium.¹⁶¹ Tracer uptake in the kidneys, which is measured at 48 to 72 hours following ⁶⁷gallium injection, is graded on a scale of 0 to 3+ and is compared with spinal intensity.¹⁸⁷ In general, a scan result of 2+ or greater renal intensity is considered positive. An animal model that examined the utility of gallium scanning in diagnosing ATIN supported this notion. ⁶⁷Gallium scanning in rats was highly accurate in differentiating experimentally induced ATIN from both drug-induced ATN and normal rat kidneys.¹⁸⁶

However, use of ⁶⁷gallium scintigraphy in humans to evaluate ATIN is marked by conflicting results. Although initial studies reflected excellent sensitivity (100%) in patients with biopsy-proven ATIN,¹⁸⁷ subsequent studies revealed lower sensitivities ranging from 58% to 69%^{188,189} with a test specificity of only 50%–60%.^{152,153,161} The major limiting factor of this modality is that positive scan results are seen with other inflammatory conditions, such as pyelonephritis, renal atheroemboli, and

glomerulonephritis, as well as noninflammatory condition such as ATN and normal kidney tissue.^{152,153,161} Renal scanning with ⁶⁷gallium scintigraphy may be useful in differentiating ATIN from ATN when kidney biopsy is high risk or is refused by the patient. However, the limitations of this imaging modality should be considered prior to using it in such patients.

FDG-PET Scanning. A noninvasive imaging test, the 2-[¹⁸F] fluoro-2-deoxy-D-glucose-positron emission tomography (FDG-PET) scan is used primarily to evaluate malignant disease but has been recently used to diagnose ATIN.¹⁶⁵ Positive FDG-PET scans were noted in two patients with severe AKI due to biopsy-proven ATIN (one of the patients had a negative gallium scan).¹⁹⁰ A third patient with severe AKI from crescentic glomerulonephritis had a negative FDG-PET scan. Following treatment and resolution of ATIN in the two patients, repeat FDG-PET scans were negative. Uptake of tracer in this setting is based on the premise that FDG accumulates in the lymphocytes, macrophages, neutrophils, and fibroblasts of inflammatory lesions.¹⁹⁰ However, this modality cannot be considered a useful test to diagnose ATIN until its true utility is substantiated in a large group of patients.

DIFFERENTIAL DIAGNOSIS

Evaluation of acute and subacute increases in serum creatinine concentration has become more standardized through the use of definitions such as the *risk-injury-failure-loss-and stage* (RIFLE), Acute Kidney Injury Network (AKIN), and KDIGO AKI criteria to diagnose and classify this entity.^{156,157,191} The KDIGO criteria include a category, AKD, that includes AKI and increases in serum creatinine or other renal-focused abnormalities that do not otherwise fit into the AKI diagnostic criteria. As previously noted, ATIN often falls into both the AKI and non-AKI AKD definitions. Regardless, these criteria do not permit differentiation of the various types of AKI or AKD, including prerenal AKI, ATN, ATIN, and other entities, which ultimately require different management approaches.

As previously discussed, ATIN may present clinically as AKI or subacute increases in serum creatinine concentration often associated with other abnormal laboratory findings, in particular urinalysis and urine microscopy abnormalities. Clinicians frequently encounter these clinical presentations in patients admitted to the general hospital wards and the intensive care units, as well as in the outpatient setting.¹⁹¹ Although the majority of hospital-acquired AKI cases are due to either prerenal AKI or ATN, unrecognized ATIN is likely the third most common cause and must be considered in the differential diagnosis.^{147,152,153} Other potential causes of an acute/subacute serum creatinine concentration rise include various forms of immune-complex and non-immune-complex glomerulonephritis, pauci-immune vasculitis, renal-limited thrombotic microangiopathies, several non-AIN tubulointerstitial diseases (renal atheroemboli, cast nephropathy, infiltrative diseases such as lymphoma and leukemia, and so on), and obstructive nephropathy. It is notable that acute or subacute increases in serum creatinine without an obvious cause can be due to biopsy-proven ATIN in a significant percentage of cases (10% to 27%).^{143,145,151–153,192,193}

Differentiating ATIN from these other causes of AKI requires a thorough clinical history, physical examination, certain laboratory data, and targeted imaging tests (Table 35.4). Ultimately, kidney biopsy to examine the underlying

histology is required to accurately make a definitive diagnosis and guide therapy. The clinical and histologic features that help distinguish between different causes of ATIN are summarized in Table 35.5.

RENAL PATHOLOGY

Diagnosis of ATIN relies on obtaining an optimal sample of kidney tissue, which includes both renal cortical and medullary tissue, via percutaneous kidney biopsy. The type of inflammatory cells, presence or absence of granuloma, interstitial edema and/or fibrosis, and other features vary based on the underlying etiology.

Tubulointerstitial Histopathology

The characteristic findings of ATIN are found predominantly in the tubulointerstitium, with the key findings of interstitial inflammation and tubulitis present. Infiltration of the renal interstitium with inflammatory cells (Fig. 35.1) is accompanied by infiltration of inflammatory cells into renal tubules, a process called “tubulitis,” and variable amounts of acute tubular injury.¹⁹⁴ In addition, there is some degree of interstitial edema and/or fibrosis. ATIN generally develops over days to weeks. When recognized and treated late, it can transition to a chronic form of interstitial nephritis, characterized by interstitial fibrosis and tubular atrophy developing over months to years.¹⁹⁴

The interstitial infiltrate is often composed of mononuclear cells, with a predominance of lymphocytes (T cells), which are primarily CD4⁺ and CD8⁺ T cells. Monocytes/macrophages, intermixed with plasma cells, B cells, small numbers of eosinophils, and possibly neutrophils are also present within the interstitium.^{194–196} The composition of the interstitial infiltrate may be helpful in determining the etiology of ATIN. For example, if a significant number of neutrophils are found within the infiltrate, the possibility of pyelonephritis (especially

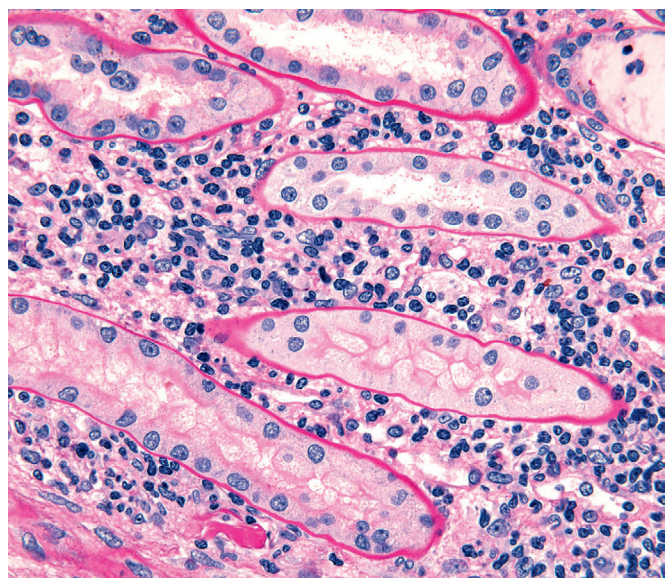


Fig. 35.1 Renal histology observed in acute tubulointerstitial nephritis highlighting the inflammatory infiltrate. Periodic acid–Schiff stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis.* 2017;24:57–63.)

Table 35.5 Clinical, Laboratory, and Histologic Features to Help in Differential Diagnosis of ATIN

Etiology of ATIN	Clinical and Laboratory Features	Distinguishing Histopathologic Features
Drug Induced	More common in older individuals	Mixed inflammatory interstitial infiltration
Proton pump inhibitors and H ₂ antagonists	Long latent period; few typical clinical features	Eosinophils may be present; rarely granulomas
Nonsteroidal antiinflammatory drugs	May be associated with nephrotic-range proteinuria, long latent period, allergic manifestations are rarely present	Eosinophils are often absent; glomerular lesions of minimal-change disease and membranous glomerulopathy rarely accompany the tubulointerstitial lesion
β -Lactams, sulfonamides, anticonvulsants	Systemic allergic manifestations of rash, eosinophilia, and fever are more common, especially with methicillin, penicillin; short latent period	Mixed inflammatory infiltrate commonly with abundant interstitial eosinophils ; tubulitis and granulomas may be present
Immune Mediated	More common in younger individuals	Mononuclear cell infiltration
SLE	Systemic features of SLE; hematuria, proteinuria, pyuria, white cell casts; SLE serologies positive; hypocomplementemia	Usually seen in association with other renal manifestations of SLE, such as focal or diffuse proliferative glomerulonephritis and “full house” immunofluorescence staining pattern; rarely seen without glomerular disease
Sarcoidosis	Hypercalcemia, hypercalciuria, urinary concentrating defects, nephrocalcinosis, nephrolithiasis, and acute or chronic kidney disease	Focal lymphocytic infiltrate and interstitial noncaseating granulomas composed of giant cells, histiocytes, and lymphocytes
Sjögren syndrome	Dry eyes and dry mouth; positive serologies	Lymphoplasmacytic infiltrate, acute tubular injury
IgG4 related	Multisystem disorder with elevated serum IgG4 level involving salivary glands, pancreas, retroperitoneum, and kidneys; affects middle-aged men	Lymphoplasmacytic interstitial infiltrate with a predominance of IgG4-positive plasma cells and interstitial fibrosis in a “storiform” pattern
Tubulointerstitial nephritis and uveitis syndrome	Uveitis characterized by painful red eyes, photophobia; common in young women	Mixed inflammatory infiltrate (including eosinophils) with noncaseating granuloma formation

ATIN, Acute tubulointerstitial nephritis; SLE, systemic lupus erythematosus.

with neutrophils forming microabscesses) or other infectious agents should be considered. An eosinophilic infiltrate is often present with drug-induced ATIN, which should be suspected when >10 eosinophils per $\times 20$ field are observed. However, eosinophilic infiltration may also be present in several other etiologies of ATIN such as tubulointerstitial nephritis and uveitis (TINU) and IgG4-related kidney disease.^{194–196} Importantly, the absence of interstitial eosinophils does not exclude a drug as causative, particularly in ATIN cases associated with PPIs and NSAIDs.

Tubulitis is diagnosed when lymphocytes are seen crossing the TBM in proximity to and crossing into/through the outer and inner aspects of the TBM.^{194–196} Physiologically this suggests extension of the interstitial inflammatory process into the tubular epithelial cells. Tubulitis is often accompanied by tubular degenerative changes, including apoptotic figures, prominent nucleoli, luminal ectasia, irregular luminal contours, cytoplasmic simplification, and loss of brush border.^{194–197} These tubular changes can be focal or diffuse and often start with denudation of the TBM as opposed to acute tubular injury, where injury often involves the villi of tubular epithelial cells with their subsequent apoptosis or necrosis.

Granulomas involving the interstitium may also be found in ATIN but is a relatively rare finding.^{198,199} Granulomas may be caseating with infectious causes, such as tuberculosis, and **noncaseating** with a number of causes, including **drugs**, **sarcoidosis** (Fig. 35.2), and idiopathic etiologies.^{198,199} Drug-induced ATIN can present pathologically as granulomatous interstitial nephritis, which is marked by the presence of

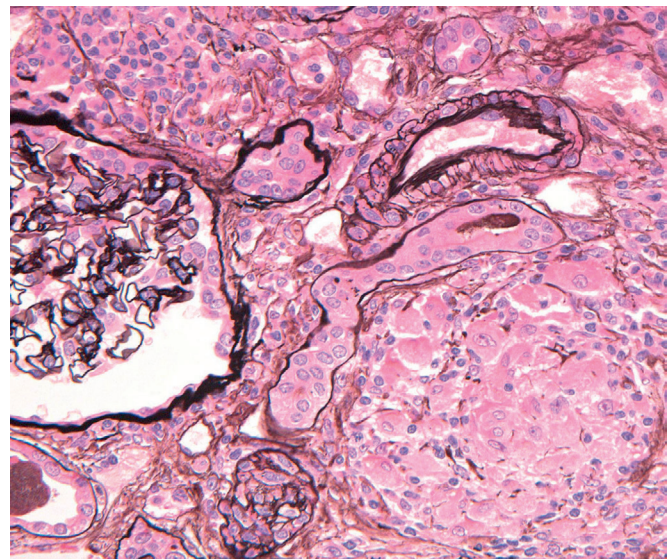


Fig. 35.2 Renal histology in sarcoidosis highlighting the inflammatory infiltrate and nonnecrotizing granuloma. Jones silver stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis*. 2017;24:57–63.)

“hypersensitivity granulomas” that are composed of reactive epithelioid histiocytes (macrophages) and multinucleated giant cells.^{198,199} Medications more commonly associated with granuloma formation include anticonvulsants, antibiotics, NSAIDs, allopurinol, checkpoint inhibitors, and diuretics.

Immunohistochemistry of renal tissue may rarely show findings that point to the pathomechanism of ATIN. Immune-complex deposits are relatively uncommon but may be present in SLE nephritis with tubulointerstitial disease and drugs associated with ATIN such as methicillin,²⁰⁰ where linear or granular staining for IgG and C3 is present on TBMs (anti-TBM antibodies). Significant IgG4 positivity in plasma cells within the interstitial infiltrate is observed with IgG4-related kidney disease.^{201–203}

Blood Vessels and Glomeruli

In the absence of underlying kidney disease, glomeruli and the renal vasculature are typically spared in ATIN. Changes related to underlying CKD, such as diabetic nephropathy, hypertension, and other renal lesions, might be seen with ATIN superimposed on these findings, especially when due to a drug.^{194–196} Kidney biopsy performed late in the course of ATIN may reveal significant interstitial fibrosis, and tubular atrophy with dilated tubular lumens. In certain forms of drug-induced ATIN, such as following NSAID exposure, glomerular lesions such as minimal-change disease or membranous nephropathy may also be observed.²⁰⁴

GENERAL TREATMENT PRINCIPLES

Treatment of ATIN hinges on identification of the underlying cause that is driving the interstitial disease. In rheumatologic disease, treatment of the underlying inflammatory condition improves kidney function. In the setting of infection-related ATIN, eradication of the infectious agent enhances kidney recovery. Early treatment is helpful in limiting chronic tubulointerstitial injury and fibrosis regardless of the cause. One of

the best examples of this is drug-induced ATIN. The mainstay of management of drug-induced ATIN is early diagnosis, followed by rapid identification of the offending agent and its prompt discontinuation.¹⁴⁴ Use of immunosuppressive agents to quell the inflammatory lesion is also an attractive option (discussed in the “Drug-Induced ATIN” section). Immunosuppressant therapy is considered standard of care for many immune disorders, including sarcoidosis, Sjögren syndrome, IgG4-related kidney disease, and TINU.^{201–203,205–209} However, although widely used for drug-induced ATIN, the supporting evidence for immunosuppressive agents is limited and controversial. In addition to these maneuvers, good supportive care is important. Blood pressure control, avoidance of nephrotoxin exposure, and management of any CKD issues improve overall patient outcomes. The following section on etiology of ATIN provides more details about specific treatment.

SPECIFIC ETIOLOGIES OF ATIN AND THEIR FEATURES

Multiple causes of ATIN exist, including drugs, infections, and systemic and idiopathic diseases (Table 35.6). Prior to the widespread use of antimicrobial agents, infections were the leading cause of ATIN. In the United States, Europe, and other developed countries, medications are now the most common cause, whereas drugs and infection are equally prevalent causes of ATIN in developing countries.²¹⁰

DRUG-INDUCED ATIN

Medications account for >70% of AIN in developed countries and approximately 50% in developing countries.^{169,210,211}

Table 35.6 Common Causes of Acute Tubulointerstitial Nephritis

Class	Specific Causative Agents
Drugs	
Antimicrobials	β -Lactams (penicillin and derivatives, cephalosporins), quinolones, ethambutol, isoniazid, macrolides, rifampin, sulfonamides, tetracycline, vancomycin, antiviral agents (acyclovir, foscarnet, indinavir, and atazanavir)
NSAIDs and COX-2 inhibitors	Almost all agents
Gastrointestinal drugs	Proton pump inhibitors, H ₂ antagonists, mesalamine, sulfasalazine
Diuretics	Furosemide, thiazides, triamterene
Anticancer agents	Ifosfamide, tyrosine kinase inhibitors, pemetrexed, immune checkpoint inhibitors
Miscellaneous	Allopurinol, amlodipine, azathioprine, captopril, carbamazepine, clofibrate, cocaine, creatine, diltiazem, phentermine, phenytoin, pranlukast, propylthiouracil, quinine, phenindione, synthetic cannabinoids
Infectious Agents	
Bacteria	<i>Corynebacterium diphtheriae</i> , <i>Escherichia coli</i> , <i>Legionella</i> , <i>Staphylococcus</i> , <i>Streptococcus</i> , <i>Yersinia</i> , <i>Brucella</i> , <i>Campylobacter</i> , <i>Legionella</i>
Viruses	Cytomegalovirus, Epstein-Barr virus, hantaviruses, hepatitis C, herpes simplex virus, HIV, polyoma virus, adenovirus
Others	<i>Leptospira</i> , <i>Mycobacterium</i> , <i>Mycoplasma</i> , <i>Chlamydia</i> , rickettsia, syphilis, toxoplasmosis, fungi
Systemic Diseases	
Immune	SLE tubulointerstitial nephritis, sarcoidosis, Sjögren syndrome, IgG4-related kidney disease, TINU, ANCA-related diseases, spontaneous tubulointerstitial nephritis
Neoplastic	Lymphoproliferative disorders, plasma cell dyscrasias

ANCA, Antineutrophil cytoplasmic antibody; COX-2, cyclooxygenase; HIV, human immunodeficiency virus; NSAIDs, nonsteroidal antiinflammatory drugs; SLE, systemic lupus erythematosus; TINU, tubulointerstitial nephritis and uveitis.

Although any medication may cause ATIN, certain medications do it frequently. Importantly, the list of medications is constantly expanding. As noted in Table 35.6, common culprits include antimicrobial agents, PPIs, NSAIDs, antiepileptics, and several anticancer drugs. A retrospective case series of 133 biopsies of ATIN noted that 70% were drug induced.¹⁶⁹ The top three drug classes identified as causes were antibiotics (49%), PPIs (14%), and NSAIDs (11%), and the top three drugs were omeprazole (12%), amoxicillin (8%), and ciprofloxacin (8%).

Antimicrobial Agents

The β -lactam antibiotics (penicillins and cephalosporins) are commonly associated with development of ATIN as part of a hypersensitivity syndrome.^{212–214} The duration of exposure to the causative β -lactam antibiotic prior to ATIN is typically relatively short, ranging from a few days to a few weeks. Fever, rash, or eosinophilia is fairly common and seen in >75% of patients. Urinary abnormalities consisting of proteinuria, pyuria, or hematuria occur in approximately 75% of affected patients. Although most patients who develop β -lactam-induced ATIN recover kidney function, CKD may also occur.^{212–214}

ATIN also occurs following exposure to non- β -lactam antibiotics such as the sulfonamide antibiotics. As with β -lactam antibiotics, these drugs may be associated with hypersensitivity reactions such as fever, rash, and eosinophilia.^{210–214} Transplant recipients, human immunodeficiency virus (HIV) patients, and CKD patients tend to develop sulfonamide-induced ATIN at higher rates than other patient groups, which likely relates to their frequent use in these populations and higher drug concentrations in patients with reduced kidney function.^{210,215,216}

The fluoroquinolone class of antibiotics may also cause ATIN. Ciprofloxacin is the most common offender, which likely relates to its widespread use. In contrast to the sulfonamide antibiotics, the fluoroquinolone antibiotics are rarely associated with a hypersensitivity syndrome. Although ciprofloxacin is the most common causative agent, ATIN does occur with other fluoroquinolone antibiotics, including norfloxacin, ofloxacin, and levofloxacin.^{210,217,218}

ATIN complicates rifampin therapy when the drug is used intermittently to treat various mycobacterial infections.^{19,23,77} In contrast to other agents, rifampin-induced ATIN is dose dependent, is associated with the production of antirifampin antibodies, and is associated with hemolytic anemia, thrombocytopenia, and hepatitis. Dialysis is required in nearly two-thirds of patients that develop rifampicin-induced ATIN.^{160,164,219}

Various antiviral agents have been associated with ATIN. However, they are rare when compared with antibiotics.^{220,221}

Proton Pump Inhibitors

PPIs are the mainstay of therapy for acid-related gastrointestinal disorders. They were first recognized as a cause of ATIN in 1992,²²² with numerous case reports and case series documenting the association.^{223–226} A retrospective study evaluated the records of two teaching hospitals in Australia over a 10-year period (1993–2003) and noted 28 cases of biopsy-proven ATIN, 18 (64%) of which were associated with PPI use.²²⁷ The mean time to development of this lesion after treatment with PPIs was 11 weeks. The Therapeutic Goods Administration database

in Australia revealed 34 cases of “biopsy-proven ATIN,” 10 cases of “suspected interstitial nephritis,” 20 cases of “unexplained renal failure,” and 26 cases of “renal impairment” associated with PPIs used over a 14-year period.²²⁷ A 2007 literature review identified 73 cases of PPI-induced ATIN, 64 of which were biopsy proven.²²⁶ All classes of PPIs were implicated, with omeprazole being the most common cause ($n = 59$).

A population-based case-control study in New Zealand patients receiving PPIs (2005–2009) examined ATIN risk that resulted in hospitalization or death.²²⁸ Current PPI users had fivefold higher odds of developing ATIN. There was a higher absolute risk in older patients ($\sim 0.2/1000$ person-years for age >60 years) compared with younger patients ($0.02/1000$ person-years for age 15–49 years). A second population-based study evaluated elderly Ontario residents who were hospitalized with AKI and ATIN within 120 days of initiating PPI therapy.²²⁹ Using a propensity score-matching algorithm, both AKI and ATIN incidence and hazard ratio (HR) were higher among patients given PPIs than among controls (AKI: 13.49 vs. 5.46/1000 person-years, HR = 2.52; ATIN: 0.32 vs. 0.11/1000 person-years, HR = 3.00). These population-based studies suggest that ATIN risk is increased in PPI users, especially the elderly population.

In contrast to ATIN induced by β -lactam antibiotics, PPI-induced ATIN is often subtle and without systemic allergic manifestations, making it challenging for clinicians to readily identify the problem. In fact, a recent study showed that only about a quarter of cases were suspected prior to a kidney biopsy.¹⁵⁰ Approximately 10% of affected patients have fever, rash, and eosinophilia, whereas <50% have fever only, <10% have rash only, and nearly 33% have eosinophilia only.²²⁶ Nonspecific complaints such as weakness, fatigue, malaise, and anorexia are the most often noted symptoms. The prognosis for PPI-induced ATIN is generally favorable with early diagnosis; patients rarely required renal replacement therapy, although CKD is a complication.²²⁷ Data suggest that clinical features and available diagnostic tests are sub-optimal for diagnosing PPI-induced ATIN and kidney biopsy is required for definitive diagnosis.

A potential long-term complication of PPI exposure is CKD. Case series and several observational studies support an association between PPI use and CKD.^{230,231} It is notable that the associations between PPI use and CKD have remained in analyses designed to account for confounding by indication, or that PPI users are sicker than the comparison group, making them more likely to experience adverse outcomes. PPI exposure leading to CKD also has biologic plausibility—unrecognized and untreated episodes of ATIN and AKI ultimately cause chronic interstitial disease and CKD.²³²

Nonsteroidal Antiinflammatory Agents

NSAIDs, including both selective and nonselective cyclooxygenase (COX) inhibitors, are widely used to treat pain, fever, and inflammation. Although the main limitations to NSAID therapy are gastrointestinal adverse effects, they are also associated with a diverse range of nephrotoxic effects with hemodynamic AKI being the most common.^{233–239} AKI developing in the setting of NSAID exposure may also occur due to ATIN. The selective COX-2 inhibitors (celecoxib and rofecoxib) have similar nephrotoxic manifestations as the nonselective NSAIDs.^{232,236,237} The clinical and pathologic findings in NSAID-induced ATIN are slightly different from

that which follows treatment with β -lactam antibiotics. Fever, rash, and eosinophilia are extremely rare, and the duration of therapy prior to the development of ATIN is long, typically in the range of 6–18 months.²⁰⁴ NSAIDs may rarely cause a dual lesion of ATIN along with a glomerular lesion (minimal change disease more common than membranous nephropathy).^{238,239}

Renal histology seen with these drugs demonstrates interstitial inflammation and tubulitis; however, they are less intense than in other forms of drug-induced ATIN and have few or no interstitial eosinophils.²⁰⁴ The clinical and pathologic differences between NSAID-induced ATIN and other agents may be due to the antiinflammatory properties of NSAIDs and possible shunting of arachidonic acid metabolites into alternative pathways that suppress immune function.⁶³ Rare cases of granulomatous ATIN and ATIN with TBM immune deposits have also been reported with NSAIDs.^{234,235} Despite severe AKI, most patients recover kidney function following drug withdrawal and, in some cases, treatment with prednisone.

Anticancer Agents

Chemotherapeutic agents are increasingly recognized as causing or contributing to the development of ATIN. A retrospective review at a single center found that nearly 5% of ATIN cases were associated with cancer chemotherapy.²⁴⁰ Among cancer patients undergoing kidney biopsy, approximately half were diagnosed with ATIN attributed to a chemotherapeutic agent. The four leading drug causes were ifosfamide (28%), bacillus Calmette–Guérin (BCG; 12.5%), tyrosine kinase inhibitors (14%), and pemetrexed (9%). It is important to remember that ifosfamide more commonly causes acute tubular necrosis and the tyrosine kinase inhibitors are more often associated with thrombotic microangiopathy.²⁴¹

A unique form of drug-induced ATIN has been described with the immune checkpoint inhibitors (CPIs).^{242–248} Tumors can survive and evade immune surveillance by overexpressing ligands that bind inhibitory T-cell receptors (immune checkpoint molecules), thereby decreasing activated T-cell infiltration into the tumor microenvironment and inhibiting antitumor T-cell responses. Two CPI drug classes use monoclonal antibodies to target immune checkpoints, which allow immune activation of T cells to invade the tumor environment and destroy cancer cells. However, activation of the immune system is associated with T-cell invasion of solid organs including the kidneys, leading to a form of autoimmune ATIN. An adverse drug effect initially noted with the CPIs was autoimmune involvement of the lung, gastrointestinal tract, and some endocrine organs. Ipilimumab, a monoclonal antibody against CTLA-4 approved to treat advanced malignant melanoma, has now been noted to cause ATIN, with or without granuloma.^{242–244} The PD-1 inhibitors (nivolumab and pembrolizumab), which are approved to treat advanced melanoma and squamous cell lung cancer, are also associated with development of ATIN.^{243–247} In a recent narrative review by the Cancer and Kidney International Network Workgroup on Immune Checkpoint Inhibitors,²⁴⁹ the incidence of AKI in patients treated with CPIs was estimated to be as high as 10%–30%. Patients who received combination therapy with ipilimumab and nivolumab had an increased incidence and severity of AKI compared with monotherapy. The onset is typically 2–3 months after initiating therapy with ipilimumab but can be delayed up to 12 months

with PD-1 inhibitors. Multiple case reports and case series suggest that steroids are an effective treatment for this complication. In the largest series of 12 patients with biopsy-proven CPI-induced ATIN, 10 received treatment with glucocorticoids, of whom 2 and 7 had complete or partial improvement in kidney function, respectively, and 2 patients were not given steroids and had no improvement in renal function.²⁴⁴

Immunosuppressive Therapy of Drug-Induced AIN

As the use of corticosteroids in the setting of drug-induced ATIN is an important and controversial topic, it is briefly reviewed. The immunologic mechanism causing ATIN and extensive inflammatory infiltrate on kidney biopsy make immunosuppressive therapy a biologically plausible treatment. However, the existing evidence is limited and derived from retrospective studies rather than randomized controlled trials.

These retrospective studies reveal **conflicting results about benefits of corticosteroid therapy in drug-induced ATIN** (Table 35.7).^{151,169,250–253} In these studies, the decision to use steroids was based on clinical judgment, which increased the risk for bias (potentially for or against steroids). Furthermore, the studies were all small and underpowered to detect any potential steroid benefit. Although these studies provide low-level evidence, they do suggest that a trial of corticosteroids in the subgroups of patients identified early (<2 weeks) without significant scarring on kidney biopsy and without significant contraindication to steroid use should be considered. Our approach is to first discontinue the offending agent. Then if kidney function does not improve within 5 days of drug discontinuation, steroid therapy should be considered. In our opinion, documentation of ATIN with kidney biopsy is preferable in all cases that are thought to be due to drugs, not only to establish the diagnosis before a potentially harmful therapy but also to evaluate the degree of scarring and possibility of response. Those responding to therapy may be continued for 4–8 weeks, whereas in the absence of response within 3–4 weeks, steroids should be promptly tapered off. A reasonable approach is to start therapy with either 3 days of intravenous pulse steroids (250–500 mg) or prednisone at 1 mg/kg. As corticosteroid therapy may not always be effective or well tolerated, other agents have rarely been used for ATIN. A 4-week trial of these agents may not be an unreasonable approach in patients where immunosuppressive therapy is deemed necessary, but glucocorticoids are not optimal.

INFECTIONS

Prior to widespread antibiotic use, streptococcus and diphtheria infections caused inflammatory reactions within the kidney. Subsequently, infections have become an uncommon cause of ATIN in the developed world, accounting for <5% of biopsy-proven cases.^{210,211} Infection may injure the tubulointerstitium either directly by renal invasion as in pyelonephritis or indirectly by an immune-mediated mechanism. Bacterial pyelonephritis should be suspected when the interstitial infiltrate consists predominantly of neutrophils and is confined to a single renal pyramid, although it becomes more diffuse when occurring in the setting of urinary obstruction. CT scanning demonstrates a wedge-shaped area of inflammation, which supports pyelonephritis rather than an allergic form of ATIN.

Table 35.7 Selected Studies of Corticosteroid Therapy in Acute Tubulointerstitial Nephritis

Author, Year	Sample Size		Peak sCr (mg/dL)		Final sCr (mg/dL)		Follow-Up (months)	Comment
	Steroid	Control	Steroid	Control	Steroid	Control		
Clarkson et al., 2004 ¹⁵¹	26	16	7.9	6.1	1.6	1.6	12	Patients received steroids late (median delay >3 weeks) after diagnosis.
Gonzalez et al., 2008 ²⁵⁰	52	9	5.9	4.9	2.1	3.7	19	Patients treated with steroids with complete recovery had shorter delay in steroid onset (13 days) as compared with those without complete recovery (34 days).
Raza et al., 2012 ²⁵¹	37	12	6.5	5.2	2.8	3.4	19	$P < .05$ for eGFR improvement for steroid versus control. No difference in renal outcomes based on steroid timing.
Muriithi et al., 2014 ¹⁶⁹	83	12	3.0	4.5	1.4	1.5	6	Steroid-treated patients had superior renal outcomes with early versus late steroid therapy.
Valluri et al., 2015 ²⁵²	73	51	4.03	3.16	NR	NR	12	Steroid-treated patients had complete recovery (48%) versus control group (41%); final sCr not significantly different at 1 year.
Prendecki et al., 2016 ²⁵³	158	29	20.5 mL/min (eGFR)	25 mL/min (eGFR)	43 mL/min (eGFR)	24 mL/min (eGFR)	24	Steroid-treated patient has significantly better eGFR at 2 years and less dialysis (5.1% vs. 24.1%). Dose, duration, and time to steroid treatment initiation were variable.

eGFR, Estimated glomerular filtration rate; NR, not reported; sCr, serum creatinine concentration.

ATIN has been also associated with a variety of other infectious agents. These include tuberculosis, leptospirosis, legionellosis, histoplasmosis, *Candida* infection, mycoplasmosis, rickettsial fever, babesiosis, and several viruses—cytomegalovirus, hantavirus, adenovirus, BK virus, and Epstein-Barr virus.^{253–265} Tuberculosis manifests as ATIN with caseating granulomas, but noncaseating granulomas can be seen.^{253,254} Histoplasmosis may present with noncaseating granulomatous interstitial nephritis.²⁶² Leptospirosis is a classic example of invasive ATIN where the spirochete enters the bloodstream through the mucosa and migrates into the tubulointerstitium, where the organism induces inflammation and tubulointerstitial injury.^{257,258} Hantavirus is associated with an inflammatory infiltrate and edema of the interstitium. Interstitial hemorrhage accompanies renal inflammation and is associated with gross or microscopic hematuria.²⁵⁵ Treatment focuses on antimicrobial treatment of underlying infection. Immunosuppressive agents are not typically used.

SYSTEMIC DISEASES

Immunologic Disorders

A variety of autoimmune disorders may cause ATIN. For the most part they are more common in the young as compared

with older patients. A number of autoimmune diseases cross over from rheumatology to nephrology due to renal involvement; those that are associated with ATIN include SLE, sarcoidosis, Sjögren syndrome, and TINU.^{211,266–272} Lupus nephritis can have ATIN alone or more commonly is associated with glomerular disease. Immune-complex deposits within the TBM drive the tubulointerstitial inflammatory lesion.

Renal sarcoidosis includes a variety of manifestations, including hypercalcemia, hypercalciuria, nephrocalcinosis, nephrolithiasis, and various glomerular lesions.^{211,266,267} ATIN is the most common intrinsic kidney lesion with sarcoidosis and is characterized by a diffuse inflammatory interstitial infiltrate that is composed of lymphocytes (T cells) sometimes with noncaseating granulomas (Fig. 35.2). Steroids are an effective therapy for sarcoidosis. However, not all cases have a complete response with many patients having partial remission and sometimes no renal improvement.^{211,266,267} As a result, CKD may be a long-term complication. Other immunosuppressive agents have been used for steroid-intolerant and steroid-dependent cases.

Idiopathic or secondary Sjögren syndrome is a systemic illness with multiorgan involvement characterized by lymphoplasmacytic infiltration.^{268,269} Although immune-complex glomerular disease rarely occurs, kidney involvement with

ATIN is far more common and ranges from 15% to 67%.^{268,269} ATIN is lymphocyte dominant and is associated with tubular atrophy, tubulitis, and variable degrees of interstitial fibrosis. Tubular involvement is clinically manifested as various tubulopathies, in particular a distal RTA. The optimal therapy for ATIN in Sjögren syndrome is unknown. Immunosuppression is generally used and may include steroids, azathioprine, cyclophosphamide, or mycophenolic acid.^{268,269}

Other immune-related disorders that primarily affect the renal microvasculature and glomeruli may also have an associated ATIN lesion.^{270,273–275} Examples include anti-glomerular basement membrane antibody disease (i.e., Goodpasture disease), immune-complex diseases (SLE or IgA nephropathy), or antineutrophil cytoplasmic antibody (ANCA)-related pauci-immune vasculitides.^{270,273–275} The presence of a concomitant active tubulointerstitial lesion with lymphocytes, plasma cells, eosinophils, and macrophages is not uncommon.

Tubulointerstitial Nephritis and Uveitis Syndrome

TINU is a rare condition of unclear etiology that presents most frequently in adolescent girls, but may appear in adulthood and rarely in the elderly.^{272,276,277} It consists of a clinical syndrome that primarily involves the eyes and kidneys.^{272,276,277} Uveitis manifests with painful red eyes and photophobia. Weight loss, fever, anemia, and hyperglobulinemia often occur before ocular and kidney manifestations. An isolated proximal tubulopathy or Fanconi syndrome can be the initial kidney manifestation.^{272,276,277} Notably, ATIN may occur before, during, or after the occurrence of uveitis. Renal histopathology reveals a mixed inflammatory infiltrate, sometimes associated with granulomas. Steroids are the mainstay of therapy for both the ocular and kidney manifestations. Therapy typically consists of 3–6 months of prednisone with a slow taper to reduce chances of relapse, which is relatively common.^{272,276,277}

Steroid-sparing options include mycophenolic acid, methotrexate, and cyclosporine. The mechanism and risk factors for this cause of ATIN are unknown; however, infections such as toxoplasmosis, Epstein-Barr infection, and giardiasis have been associated with TINU. No clear-cut immune or genetic cause has been described. Fortunately, treated patients have a relatively good prognosis.

IgG4-Related Kidney Disease

IgG4-related kidney disease is part of a relatively recently recognized group of disorders characterized by elevated serum levels of IgG4 subclass antibodies (60%), tissue deposition of IgG4, and involvement of various organ systems, including salivary glands, pancreas, retroperitoneum, and kidneys.^{278–282} Middle-aged men most commonly develop this disease. IgG4-related kidney disease can be very difficult to diagnose unless the patient develops a renal lesion in the setting of known or suspected IgG4-related systemic disease. Kidney involvement from this disease is fairly wide ranging and includes acute or chronic interstitial nephritis, granulomatous ATIN, masslike kidney lesions, and obstructive nephropathy due to retroperitoneal fibrosis, which may present in the absence of hydronephrosis on imaging.^{278–282} A lymphoplasmacytic interstitial infiltrate with a predominance of IgG4-positive plasma cells is observed on histology. Interstitial fibrosis with a “storiform” pattern that is typified by a cartwheel appearance of arranged fibroblasts and inflammatory cells

are somewhat characteristic of this disease.^{278–282} Therapy consists of long-term glucocorticoid therapy; however, steroid-sparing agents such as mycophenolic acid or rituximab have been used with some success.

PROGNOSIS

The prognosis of ATIN is best characterized for drug-induced disease, which is generally considered reversible with early recognition and withdrawal of the offending agent. Patients with drug discontinuation within 2 weeks of disease onset are more likely to recover kidney function than those who remain on the precipitating medication for ≥ 3 weeks. However, renal recovery after an episode of ATIN is not complete in over half of the patients, and studies with long-term follow-up have found progression of AIN to CKD in 70%–88%.^{175,283,284} In a review of three ATIN case series, it was noted that only 64.1% of patients made a full recovery, whereas 23.4% gained a partial recovery, and 12.5% developed CKD and remained on renal replacement therapy.¹⁵⁸ Worse prognosis is associated with increasing age, but there appears to be no correlation with severity of AKI. The severity of tubulointerstitial disease (in particular, focal vs. diffuse interstitial cellular infiltration) is not associated with outcome.²⁸⁵

CHRONIC TUBULOINTERSTITIAL DISEASES

CTIN is characterized by progressive, often indolent, loss of kidney function along with the pathologic features of tubular atrophy, interstitial mononuclear cell infiltrates, and interstitial fibrosis. CTIN has numerous etiologies (Table 35.8) with common clinical features

CLINICAL FEATURES

Patients with chronic tubulointerstitial disease are brought to medical attention from either the findings of abnormal laboratory testing (elevated serum creatinine or abnormal urinalysis) or symptoms due to a systemic disease (sickle cell anemia). Depending on the level of kidney disease at the time of presentation, patients may complain of nonspecific symptoms attributable to CKD, such as fatigue, weakness, decreased appetite, weight loss, and others. Because chronic tubulointerstitial diseases may lead to decreased ability to concentrate the urine, patients may complain of nocturia. Typical laboratory findings in these patients included nonnephrotic-range proteinuria that is typically low-molecular-weight (tubular) in nature (lysozyme, $\beta 2$ -MG, NAG, and retinol-binding protein), microscopic hematuria and pyuria (including leukocyte casts), glycosuria (due to tubular defects), and if associated with an infectious etiology, positive urine cultures. Some causes of CTIN display characteristic patterns of tubular dysfunction. Proximal tubular defects may include glycosuria, bicarbonaturia, phosphaturia, aminoaciduria, and a proximal RTA. Distal tubular defects may include hyperkalemia, sodium wasting, and a distal RTA. Medullary injury may lead to sodium wasting and urinary concentrating defects. Serum uric acid levels are usually lower than expected for the degree of renal failure, presumably because of tubular defects in the reabsorption of uric acid. Anemia occurs relatively early in the course of certain forms of chronic interstitial disease, presumably because of early

Table 35.8 Etiologies of Chronic Tubulointerstitial Nephritis

Drugs and Toxins	Metabolic Disorders	Immune Mediated	Infection	Hematologic	Others
Combination analgesics	Abnormal uric acid metabolism	Sarcoidosis	Bacterial pyelonephritis	Sickle cell disease	Radiation nephritis
5-Aminosalicylic acid		Behçet syndrome	Hantavirus	Light-chain nephropathy	Secondary to progressive glomerular disease
NSAIDs	Hypokalemia	Sjögren syndrome	Leptospirosis	Amyloidosis	Obstruction
Chinese herbs	Hypercalcemia	Inflammatory bowel disease	Xanthogranulomatous pyelonephritis	Multiple myeloma	Vesicoureteral reflux
Lithium	Hyperoxaluria	TINU	Malacoplakia		
Lead	Cystinosis	IgG4-related systemic disease			
Cadmium		Allograft rejection			
Mercury		Systemic lupus erythematosus			
Balkan endemic nephropathy					
Calcineurin inhibitors					
Cisplatin					
Herbicides (glyphosate)					
CKD undetermined origin					

※XPN: *E.Coli*, *Proteus mirabilis*

CKD, Chronic kidney disease; Ig, immunoglobulin; NSAID, nonsteroidal antiinflammatory drug; TINU, tubulointerstitial nephritis and uveitis.

destruction of erythropoietin-producing interstitial cells. Ultrasonography is typically normal in most patients with CTIN until late in the course when the kidneys appear echogenic and shrunken. In some cases, such as CTIN due to analgesic toxicity, the kidney may have irregular contours and calcifications.

DIFFERENTIAL DIAGNOSIS

Patients suspected of having CTIN usually present with CKD that is progressive and often indolent in nature. The urinalysis may show nonnephrotic-range proteinuria and thus many of these patients may not undergo diagnostic renal biopsy. Therefore a high degree of suspicion for the diagnosis is required. As shown in Table 35.8, there are a myriad of etiologies for CTIN and a careful history and physical examination may uncover clues as to the cause. For instance, exposure to heavy metals, heavy NSAID use, other drug or toxin exposure, chronic infections, or other chronic systemic diseases may serve to form a narrow differential diagnosis that can be supplemented by appropriately focused laboratory testing (such as heavy-metal levels, immunologic testing, or others). Kidney biopsy can be especially useful when CTIN is secondary to immune-mediated, paraprotein-associated, or infectious etiologies.

RENAL PATHOLOGY

Pathologic features of CTIN are largely conserved and stereotyped across a wide variety of distinct causes. These features include atrophy of tubular cells with flattened epithelial cells and tubule dilation, interstitial fibrosis, and areas of mononuclear cell infiltration within the interstitial compartment and between tubules. TBMs are frequently thickened. The cellular infiltrate in chronic tubulointerstitial disease is composed of lymphocytes, macrophages, and B cells, with only occasional neutrophils, plasma cells, and eosinophils. This infiltrate is typically less marked than in ATIN. If immunofluorescent studies are performed on biopsy specimens, they might occasionally reveal nonspecific

immunoglobulin or C3 deposition along the TBMs. As described later, specific etiologies of chronic tubulointerstitial disease may show additional characteristic findings on renal biopsy.

In chronic tubulointerstitial disease, glomeruli may remain remarkably normal by light microscopy, even when marked functional impairment is present. As chronic interstitial injury progresses, glomerular abnormalities become more evident and consist of periglomerular fibrosis, segmental sclerosis and, ultimately, global sclerosis. Small arteries and arterioles show fibrointimal thickening of variable severity.

GENERAL TREATMENT PRINCIPLES

Progressive CKD associated with CTIN is best approached by addressing any underlying systemic disease (such as appropriate immunosuppressant therapy for sarcoidosis), removing and avoiding any drug or toxin exposure, and addressing any other reversible condition that may hasten progression (such as diabetes mellitus or hypertension). Otherwise, treatment largely follows existing paradigms for the care of patients with CKD²⁸⁶ (see Chapter 59: Staging and Management of Chronic Kidney Disease). An exception to this is that there are no data supporting a protective role for RAAS blockade in CTIN as there is for patients with proteinuria or glomerular-predominant diseases.

SPECIFIC ETIOLOGIES OF CHRONIC TUBULOINTERSTITIAL NEPHRITIS

Table 35.8 lists the etiologies associated with CTIN and selected causes are discussed here.

ANALGESIC-INDUCED CHRONIC TUBULOINTERSTITIAL NEPHRITIS (ANALGESIC NEPHROPATHY)

Long-term (years) ingestion of large quantities of combination analgesic medications has been associated in epidemiologic studies with chronic interstitial nephritis (termed analgesic nephropathy) and papillary necrosis. The incidence of analgesic nephropathy varies among different countries and

among different U.S. geographic areas and is highly dependent on the period sampled. Before the removal of phenacetin (between 1960 and 1980) from analgesic mixtures, analgesic nephropathy had been reported as a common cause of CKD in Scotland, Belgium, and Australia, accounting for 10% to 20% of patients with ESRD in those countries.²⁸⁷ In the United States, case-control studies support the conclusion that variations in the frequency of analgesic nephropathy track with patterns and the degree of analgesic use.²⁸⁸ In the 1990s, there was a decrease in the prevalence and incidence of analgesic nephropathy among patients undergoing dialysis in several European countries and Australia. Most authors associated this decrease with the removal of phenacetin from analgesic mixtures.²⁸⁹ The true prevalence of analgesic nephropathy is not known, in part because definitive diagnostic criteria do not exist. The condition is often a diagnosis of exclusion in CKD patients: a history of heavy analgesic use and no other etiologies that can explain their progressive kidney disease. No histologic features are characteristic of analgesic nephropathy.

Analgesics implicated in analgesic nephropathy are compound mixtures that contain aspirin or antipyrine in combination with phenacetin, acetaminophen (paracetamol), or salicylamide and caffeine or codeine in over-the-counter proprietary mixtures. Development of analgesic nephropathy likely requires prolonged regular ingestion of combination analgesics (at least 6 tablets daily for >3 years). Despite numerous observational studies, the relative contribution of specific drugs to the development of analgesic nephropathy is not clear and often data on individual components of analgesic combinations do not show nephrotoxicity.²⁹⁰

Data are strongest for phenacetin, an analgesic antipyretic compound that is metabolized to paracetamol and other reactive intermediates. Given the concern for renal toxicity, phenacetin has been withdrawn from the market in most countries. Supporting its association with renal toxicity, there was a marked reduction in analgesic nephropathy incidence in those countries, coinciding with this withdrawal.²⁸⁹ Interestingly, phenacetin compounds remain in use in some countries, and recent Chinese data suggest that phenacetin users are more likely to have CKD than a control population of nonusers.²⁹¹ However, this study requires cautious interpretation because, after adjusting for confounders for the development of CKD, there was no statistically significant effect. It is also important to note that the risk for analgesic nephropathy with phenacetin occurs solely in compound mixtures and that experimental rat studies of long-term phenacetin monotherapy at therapeutic doses have not demonstrated nephrotoxicity, although chronic suprathreshold administration has repeatedly been found to produce papillary necrosis.²⁹²

Paracetamol (acetaminophen) has also been recognized as a potential cause of analgesic nephropathy. In one prospective study, lifetime intake of >500 g of paracetamol was associated with an approximately twofold higher risk of developing a GFR decline to < 30 mL/min 1.73 m².²⁹³ However, direct causation has not been demonstrated and several studies have not demonstrated an association between CKD and paracetamol use.²⁹⁴

There are few data to suggest that long-term aspirin monotherapy is associated with the development of analgesic nephropathy.^{295,296} In a large case-control study of 11,032 men, there was no association between serum creatinine

levels and aspirin consumption.²⁹⁵ A more recent cohort study also failed to find an association with long-term aspirin monotherapy with CKD.²⁹⁶ However, mixtures containing both paracetamol and aspirin are associated with greater renal toxicity than either agent alone.^{297,298} For instance, a Swedish study demonstrated a twofold higher incidence of raised creatinine levels in users of paracetamol-aspirin as compared with users of aspirin alone.²⁹⁸

The majority of studies have failed to find an association between heavy NSAID use and CKD.^{293,295} However, chronic repeated intake of NSAIDs may increase the rate of progression of CKD in those patients with preexisting renal impairment.²⁹⁹

Analgesic nephropathy is recognized far more frequently in women than in men (five to seven times). The patients typically give a history of chronic headaches, joint pain, and/or abdominal pain that prompts regular use of analgesics. An episode of flank pain with or without associated hematuria may indicate a sloughed and potentially obstructing papilla. Because these drugs are available over the counter, many patients may not come to the attention of healthcare professionals until kidney disease has reached an advanced stage. At that point, renal functional abnormalities attributable to CTIN are nonspecific, including nocturia, sterile pyuria, and azotemia. Anemia is common. Discontinuation of heavy analgesic use can slow or arrest progression of the renal disease, but reversal of the kidney dysfunction is unlikely.

Mechanistically, analgesic components such as acetaminophen can injure cells through lipid peroxidation.³⁰⁰ These drugs and metabolites are present in highest concentration in the medulla and papillary tip, which is where the initial lesions of capillary sclerosis are seen.³⁰¹ In the presence of aspirin, there is competition for glutathione within the cortex and papillae of the kidney. If cellular glutathione is depleted, there is the possibility of potentiation of the renal toxicity of acetaminophen, and its reactive metabolites.³⁰¹ In addition, because aspirin and other NSAIDs can suppress the production of vasodilatory prostaglandins, renal blood flow to the medulla may be compromised, adding a hemodynamic contribution to injury.

Imaging findings may be suggestive but not diagnostic for analgesic nephropathy. Findings on noncontrast CT scans of small kidneys, with bumpy renal contours and papillary calcifications, were found to diagnose analgesic nephropathy with greater sensitivity and specificity than clinical signs and symptoms (Fig. 35.3).³⁰² However, these CT findings are infrequent in the U.S. ESRD population and do not occur frequently enough among those patients with heavy and sustained analgesic use to make it a sensitive tool to detect analgesic nephropathy.³⁰³

The late course of analgesic nephropathy may be complicated by urinary tract malignancy.³⁰⁴ The major presenting symptom of this complication is microscopic or gross hematuria. It has been estimated that a urinary tract malignancy will develop in as many as 8% to 10% of patients with analgesic nephropathy, typically after 15 to 25 years of heavy analgesic use.³⁰⁵ Some of these patients may not have been previously diagnosed with analgesic nephropathy.³⁰⁶ Pathogenesis of these uroepithelial malignancies presumably relates to the concentration and accumulation of phenacetin metabolites with alkylating capabilities within the renal medulla and lower urinary tract.³⁰⁷ Whether NSAIDs and

Analgesic nephropathy (AN)

Macroscopic aspect of an AN kidney

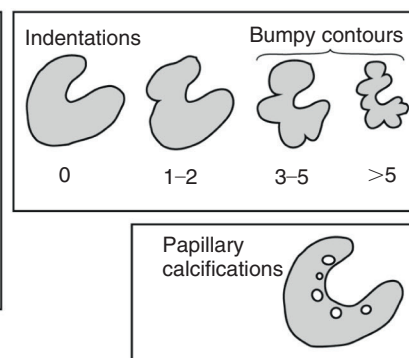
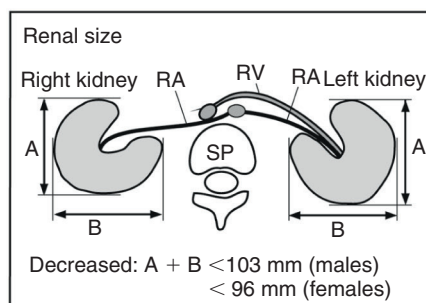
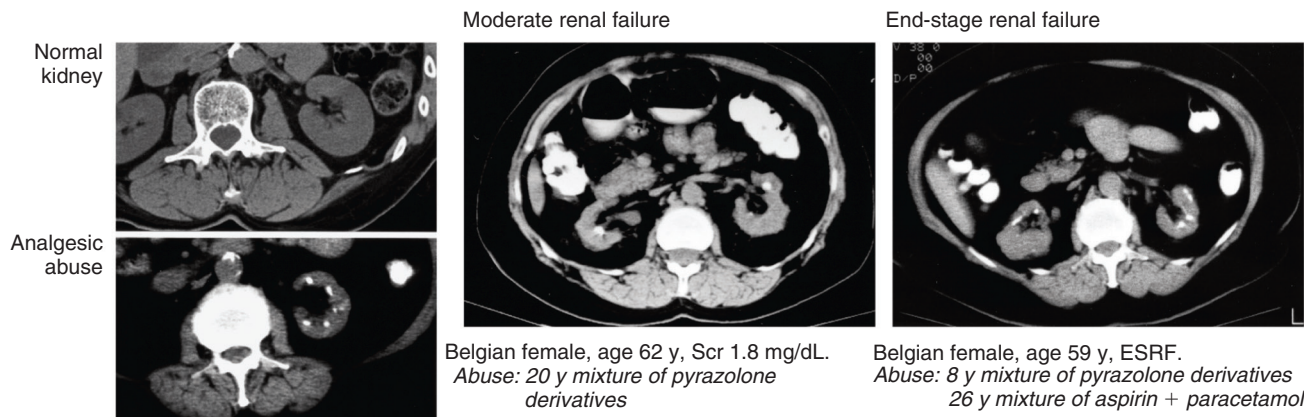
**Measurement of diagnostic criteria****CT scans without contrast material**

Fig. 35.3 Renal imaging criteria of analgesic nephropathy (AN) as observed in a postmortem kidney and in computed tomography (CT) scans without contrast material. These criteria include a decreased renal size, bumpy contours, and papillary calcifications. RA, Renal artery; RV, renal vein; SP, spine. (Adapted from De Broe ME, Elsevier MM. Analgesic nephropathy. *N Engl J Med.* 1998;338:446–452.)

acetaminophen are associated with these kidney tumors remains controversial. However, a recent meta-analysis of epidemiologic studies has suggested that both classes of drugs are associated with a significant increase in the risk of developing kidney cancer, regardless of the presence of analgesic nephropathy.³⁰⁸

ARISTOLOCHIC ACID-INDUCED INTERSTITIAL NEPHRITIS (BALKAN AND CHINESE HERB NEPHROPATHY)

Aristolochic acids (AAs) are compounds derived from plants of the genus *Aristolochia* and have been used in various manners ranging from antiviral and antibacterial agents to components of weight-loss preparations.³⁰⁹ Of note, two species, *Aristolochia clematitis* and *Aristolochia fangchi*, with common names of birthwort and Fang chi, have been identified as compounds responsible for kidney injury leading to progressive renal failure in a syndrome that was originally referred to as “Chinese herb nephropathy,” but later became known as AA nephropathy (AAN).³¹⁰

In the early 1990s, reports from Belgium described patients, frequently women, presenting with a rapidly progressive form of renal failure. Kidney biopsies from these patients revealed findings consistent with those of CTIN.³¹¹ Initial reports noted that affected patients shared a history of chronic ingestion of the same preparation of Chinese slimming herbs as part of a weight-loss regimen. The number of case reports grew throughout the 1990s to over 120 cases by early 2000.³¹⁰ Cases

were described throughout the world. Investigations into the cause of the initial outbreak demonstrated that an herb with no known nephrotoxicity (*Stephania tetrandra*) had been substituted with *A. fangchi*, which was toxic. Subsequent studies demonstrated several clinical subtypes, including AKI, tubular dysfunction with normal serum creatinine levels, and CTIN (the most common presentation).³¹⁰ These subtypes likely represent clinical variants that varied due to the amount and duration of exposure to AA. For example, the CTIN variant was seen more commonly in those patients with low daily intake but large cumulative doses.

Patients with AAN usually present with elevated serum creatinine values and few other symptoms. Other presenting signs and symptoms of AAN include anemia out of proportion to the degree of kidney failure (destruction of peritubular erythropoietin-producing cells), glucosuria, mild tubular proteinuria, and sterile pyuria. Renal ultrasound, especially when CKD is advanced, shows small kidneys with irregular contours.³¹² Diagnostic criteria for AAN have been proposed and include an estimated GFR (eGFR) < 60 mL/min/1.73 m², and two of the three following histopathologic criteria: (1) hypocellular interstitial fibrosis decreasing from the outer to the inner cortical regions; (2) ingestion of AA-containing product, confirmed by phytochemical analysis; and (3) presence of DNA adducts in the renal tissue or urinary tract that can be identified in specialized laboratories.³¹³

Evidence from clinical investigation and animal models has suggested that AA is the primary etiologic agent of these

original presentations as well as having a significant contribution to the epidemiology of CKD and ESRD in Taiwan and mainland China where these herbal preparations are ingested.^{314,315} AA now appears to be the common factor underlying the development of the nephropathy associated with Chinese herbs and Balkan nephropathy (see later).³¹⁶ The role of AA in producing this lesion has been supported by the creation of a similar renal lesion in rabbits and rats to which AA was administered regularly over weeks to months.^{317,318} Kidneys from these animals display interstitial fibrosis, tubular atrophy, and some cellular infiltration, along with some atypical and/or malignant uroepithelial cells. Given the role of AA in causing nephropathy as well as uroepithelial malignancies, most health agencies worldwide have enacted bans on supplements containing this compound. However, due to lack of regulatory oversight over supplements, many AA-containing products can still be found for sale.³¹⁹

Several therapies have been tried to slow the progression of AAN to ESRD.³¹⁰ Obviously, avoidance of further AA exposure is critical. However, the only therapy that has demonstrated possible benefit is glucocorticoids, which has been investigated in observational studies in very small numbers of patients and appears to delay progression of CKD.^{320,321} Given the lack of other effective therapies, it is reasonable to consider a course of glucocorticoids in patients with documented AAN.

The potential for uroepithelial malignancies in patients exposed to AA is well documented. For example, in a study of 39 patients with Chinese herbal nephropathy and ESRD who underwent removal of their native kidneys and ureters, urothelial carcinoma was discovered in 18 patients and mild-to-moderate urothelial dysplasia in 19 patients.³²² The mutational footprint induced by AA has been studied and the average mutation rate induced by this compound is nearly 20-fold higher than that seen in lung cancers associated with smoking.³²³ This makes AA one of the most potent carcinogens known to man. Such findings have led to recommendations that patients with a history of AA exposure should undergo regular cancer surveillance with yearly CT, cystoscopy with ureteroscopy, and biannual urine cytology.^{310,312} When renal replacement therapy is started and prior to transplantation, strong consideration should be given for bilateral nephroureterectomy.

Balkan endemic nephropathy (BEN) is a chronic tubulointerstitial disease seen largely in families residing on the alluvial plains of the Danube River in Serbia, Bosnia, Herzegovina, Croatia, Romania, and Bulgaria. The prevalence is high in these areas, ranging from 0.5% to 5%.³²⁴ Although genetic and environmental factors were suspected to play a role in its pathogenesis, current evidence suggests that AA may be the primary etiologic agent.³¹⁰ BEN may result from the ingestion of bread contaminated with seeds of *A. clematitis*, which leads to a low-grade but extended exposure to AA. This mechanism is supported by the finding of AADNA adducts in kidney and urothelial tumor specimens from BEN patients.³²⁵

Patients with BEN share nonspecific clinical features common to many chronic tubulointerstitial diseases. Renal and urinary abnormalities first appear after residing in an endemic area for at least 15 years. Affected individuals are not usually hypertensive. Initial renal abnormalities include tubular dysfunction characterized by tubular proteinuria,

glycosuria, aminoaciduria, and impaired acid excretion. On kidney biopsy, there is a hypocellular infiltrate and tubulointerstitial fibrosis that decreases from the outer to inner cortex. This is temporally followed by impaired concentrating ability and slowly progressive azotemia, which can result in ESRD over a period of >20 years (unlike AAN, which progresses at a faster pace). A normochromic normocytic anemia is typical as well. Kidneys are of normal size early in the course of the disease, but kidney size diminishes with time. As with AAN, there is a strong association of BEN with urothelial carcinoma and thus screening for these cancers is critical in exposed individuals.³²⁵ Diagnosis is presumptive, based on renal abnormalities consistent with chronic tubulointerstitial disease in a patient residing in an endemic area.³²⁶ There is no specific therapy for this form of tubulointerstitial disease.

HEAVY-METAL-ASSOCIATED CHRONIC TUBULOINTERSTITIAL NEPHRITIS

Occupational or environmental exposure to heavy metals has been implicated in the etiology of CKD. The linkage between heavy metals and nephrotoxicity is strongest for lead and cadmium. Although arsenic and uranium exposure has been hypothesized as nephrotoxins, the data in this regard are weak.^{327,328}

Lead

Epidemiologic studies have strongly implicated excessive exposure to lead as a cause of CTIN leading to renal failure.³²⁹ Although occupational and environmental exposure to lead has significantly decreased, continued exposure to low levels of lead occurs through old water pipes, pottery, crystal, and lead-based paint in older dwellings. Most recently, lead exposures in larger amounts have occurred in contaminated water supplies such as in Flint, Michigan, in 2014. Given the prevalence of low-level lead exposure, it is critical to understand whether this exposure contributes, solely or in combination with other factors to the development of CKD. The determination of lead exposure requires a combination of careful history taking as well as the accurate measurement of body lead burden, which is accomplished through the ethylenediaminetetraacetic acid (EDTA) mobilization test. This test is required because blood lead levels only indicate recent, not chronic, lead exposure. Several studies have demonstrated correlations between elevated blood levels and/or body lead burden with the presence of kidney disease and/or accelerated rates of progression of chronic renal disease.^{330,331} Using data from the Third National Health and Nutrition Examination Survey (NHANES III), Muntner and colleagues concluded that among the adult U.S. population with hypertension, even low-level exposure to lead is associated with CKD.³³¹ The same study also showed that in U.S. adolescents, higher blood lead levels correlated with lower GFRs.³³² Animal studies have also supported the hypothesis that ingestion of lead can accelerate CKD.³³³

The pathogenesis of chronic lead nephropathy is related to the reabsorption of filtered lead by the proximal tubule leading to defects in proximal tubule function, especially in children, including aminoaciduria, glycosuria, and phosphaturia representing Fanconi syndrome. Renal biopsies in adult patients with subclinical lead nephropathy and a mild-to-moderate decrease in GFR primarily show chronic interstitial nephritis, tubular atrophy, and interstitial fibrosis.

Adults who develop CTIN in association with lead exposure typically have hypertension and frequently gout as well. Thus the combination of progressive CKD, gout, and hypertension should prompt at least consideration of possible lead exposure. Progression to ESRD develops over years. Such slow progression has encouraged clinical investigators to examine whether chelation of lead might slow or reverse the progression of this disease. Earlier studies with small numbers of patients suggested that chronic injections of EDTA in patients with mild renal insufficiency and industrial exposure to lead could improve GFR.³³⁴ Two larger prospective studies in patients with nondiabetic CKD, no lead exposure history, and normal or low-normal total body lead burdens demonstrated that chronic EDTA chelation can improve GFR over a 24-month period relative to the control group.^{335,336} Although EDTA chelation may be exerting a benefit through processes other than lead removal, it is intriguing to consider that lead may adversely affect the progression of other forms of renal disease and that this may be positively affected by EDTA chelation. Use of EDTA chelation therapeutically for chronic lead nephropathy still requires confirmation in larger studies.

Cadmium

Cadmium accumulates in the body after gastrointestinal absorption or inhalation. Cadmium nephropathy can develop in those with prolonged low-level exposure to excess cadmium, such as with zinc smelter workers, or in the setting of massive environmental contamination, as occurred in Japan in the early part of the 20th century. Prominent features of this poisoning included renal failure.³³⁷ Cadmium is bound to metallothionein, and proximal tubular cells take up the complexes. Cadmium induces a tubular proteinuria.³³⁸ Other proximal tubule defects, including renal glycosuria, aminoaciduria, hypercalciuria, and phosphaturia, are also observed. Uncommonly, such renal damage may progress to an irreversible reduction in glomerular filtration.³³⁹

High levels of environmental contamination with cadmium related to mining in Japan were an unusual event. The extent to which chronic, low-level environmental exposure to cadmium affects renal function is much less clear. The Cadmibel Study recruited over 2000 adults from different areas of Belgium in an attempt to compare the relationships among hypertension, cardiovascular disease, and renal abnormalities with urinary cadmium excretion, a measure of lifetime cadmium exposure. The results demonstrated that although hypertension and cardiovascular risk were not associated with urinary cadmium excretion, there was a direct correlation of urinary cadmium excretion with serum alkaline phosphatase activity, and with urinary excretion of retinol-binding protein, NAG, β 2-MG, amino acids, and calcium, suggesting that cadmium exposure is associated with renal tubular injury.³³⁹ Interestingly, 5-year follow-up study of individuals with the highest urinary cadmium excretion did not demonstrate evidence of progressive kidney damage or loss of function.³⁴⁰ Increased urinary calcium excretion seen in those exposed to cadmium has been associated with calcium phosphate kidney stones.³⁴¹

LITHIUM-INDUCED CHRONIC TUBULOINTERSTITIAL NEPHRITIS

Lithium has multiple nephrotoxic effects ranging from nephrogenic diabetes insipidus associated with diminished

expression of the vasopressin-regulated water channel aquaporin-2 to progressive CKD from CTIN.^{342–344} Animal models of chronic lithium ingestion reveal the development of focal interstitial nephritis, including tubular atrophy, interstitial fibrosis, and distal tubular dilation and microcyst formation. Glomerular sclerosis and azotemia also occurred by 12 months in these animals.³⁴⁵ In a study of patients treated chronically with lithium (mean duration of therapy > 13 years), kidney biopsies from 24 patients with abnormal renal function were studied.³⁴⁶ In this selected population, all the patients had findings consistent with CTIN on biopsy. Tubular dilation and cysts were also present in the cortex and medulla (Fig. 35.4). Development of these cysts is likely a direct consequence of lithium uptake by principal cells, because these cysts mainly originate from the distal tubule and collecting duct. A number of patients also had focal or global glomerulosclerosis. A serum creatinine value >2.5 mg/dL at the time of biopsy was the most powerful predictor of ultimate progression to ESRD, which occurred even if lithium was discontinued. Approximately 20% of long-term lithium patients demonstrated slowly rising serum creatinine values,

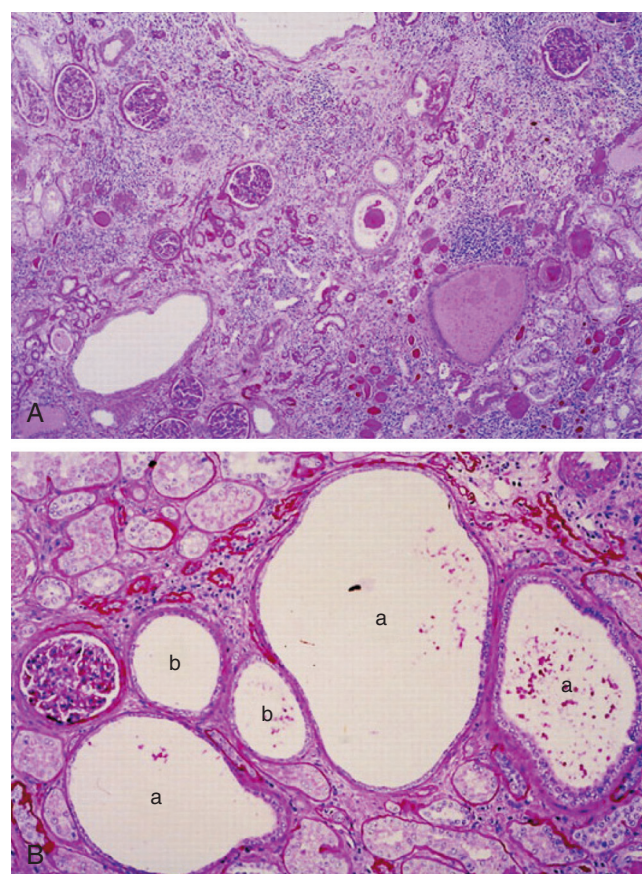


Fig. 35.4 (A) Severe lithium-associated chronic tubulointerstitial nephropathy, with the additional finding of focal tubular cysts arising in a background of severe interstitial fibrosis and tubular atrophy. (Periodic acid–Schiff [PAS] stain, $\times 40$.) (B) High-power view of tubular cysts lined by simple cuboidal epithelium (a). Adjacent tubules show tubular dilation (b). (PAS stain, $\times 100$.) (From Markowitz GS, Radhakrishnan J, Kambham N, et al. Lithium nephrotoxicity: a progressive combined glomerular and tubulointerstitial nephropathy. *J Am Soc Nephrol*. 2000; 11:1439–1448.)

but this did not correlate with duration of therapy or cumulative dose.³⁴⁷ Another study from Sweden has described a sixfold increase in prevalence for ESRD in patients taking lithium compared with the general population. Prevalence of CKD in the lithium-treated population was about 1.2%, with duration of therapy as the only identified risk factor.³⁴⁸ Molecular alterations associated with lithium use are multiple and include changes in calcium signaling, inositol mono-phosphate, extracellular-regulated prostaglandin, sodium/solute transport, G-protein–coupled receptors, nitric oxide, vasopressin/aquaporin, and inflammation-related pathways.³⁴⁹ How these may interact to lead to tubulointerstitial injury and fibrosis is not clearly elucidated, but elevated production of TGF- β 1 may be a final common pathway leading to fibrosis.³⁵⁰

Clinical management of patients on chronic lithium who develop CKD requires judgment and a risk–benefit analysis in conjunction with a discussion of possible alternative medications that can be used to treat the patient. Lithium levels should be carefully monitored and maintained at the lowest level that controls symptoms. Given the more recent development of additional drugs (olanzapine, quetiapine, lamotrigine) for first-line treatment of bipolar disorder, it is reasonable to use these drugs in patients with an elevated creatinine level, preferably before irreversible interstitial damage has occurred. Although experimental and clinical evidence has demonstrated the efficacy of the epithelial sodium channel inhibitor amiloride for the treatment of lithium-induced nephrogenic insipidus, it is not known whether this drug can prevent CKD.³⁵¹

URIC ACID NEPHROPATHY

As the major organ responsible for the excretion of uric acid, the kidney can be affected in a number of ways by abnormal uric acid metabolism. In general, problems arise following uric acid crystallization in the tubules, collecting system, or outflow tract or the deposition of uric acid within the interstitium, with resulting inflammation. Uric acid solubility is pH and concentration dependent. Thus because uric acid in the tubular lumen is concentrated and exposed to a lower pH in the distal nephron, the likelihood of precipitation increases in these distal segments. Uric acid stones are a well-recognized entity and are discussed in Chapter 38: Urinary Stone Disease. Acute uric acid nephropathy, typically seen as an AKI phenotype following cell breakdown with tumor lysis syndrome, is also a recognized entity for which prophylactic treatment (rasburicase and allopurinol) can usually be given.

Whether sustained and chronic hyperuricemia leads to CTIN has been a controversial issue. Claims in the 1970s that up to 11% of chronic interstitial disease could be attributed to disorders of uric acid metabolism were challenged in the 1980s because of the difficulty in identifying effects of hyperuricemia independent of other risk factors, such as hypertension, vascular disease, kidney stones, or aging.^{352,353} An additional controversy was related to the possibility that patients with coexisting gout and CTIN might have chronic lead intoxication as a cause of both disturbances. Recently, several studies have confirmed an association between hyperuricemia and the development and progression of CKD. In a study of >13,000 participants followed for up to 9 years, every 1-mg/dL increase in baseline uric acid level

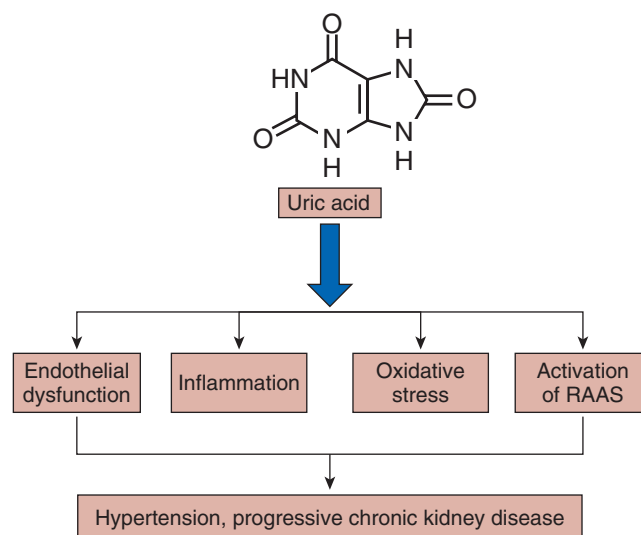


Fig. 35.5 Pathogenesis of uric acid–associated chronic kidney disease. RAAS, Renin–angiotensin–aldosterone system.

was associated with a 7% increased risk for developing kidney disease after adjustment for confounders.³⁵⁴ A meta-analysis of 13 observational studies also identified hyperuricemia being associated with an increased risk of new-onset CKD.³⁵⁵ Other studies have documented that hyperuricemia is also associated with an increased risk of progression to ESRD.^{356,357}

Given these associations, a few small trials have assessed the effect of uric acid-lowering therapies (allopurinol, febuxostat) on the progression of CKD. Although small, these studies have shown promising results with slower progression of CKD in patients treated to lower uric acid levels.^{358–361} Losartan also has the ability to lower uric acid levels by decreasing uric acid absorption in the proximal tubule. In the Reduction of Endpoints in Non-insulin diabetes mellitus with Angiotensin II Antagonist Losartan (RENAAL) trial, the risk of renal events was decreased by 6% for every 0.5-mg/dL decrease in uric acid levels.³⁶² Larger-scale randomized trials are needed to confirm these results and to inform clinical practice. However, it seems reasonable to recommend dietary restriction of protein and purines in patients with gout and interstitial disease or patients with CKD and an elevated serum urate level.

Mechanistically, hyperuricemia may contribute to progressive CKD through several overlapping mechanisms that lead to endothelial dysfunction, inflammation, oxidative stress, and activation of the renin–angiotensin system (Fig. 35.5).³⁶³ Furthermore, uric acid regulates critical proinflammatory pathways in vascular smooth muscle cells, potentially having a role in the vascular changes associated with hypertension and vascular disease.^{364,365} Animal studies have also revealed that hyperuricemia induces hypertension and accelerates renal progression in the remnant kidney model.^{366,367} Taken in aggregate, these studies provide evidence that uric acid may be a true mediator of kidney disease and progression.

CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

In 2002, a cohort of patients with advanced CKD without a clear etiology was observed in El Salvador.³⁶⁸ Subsequently,

several other epidemiologic studies from the coastal regions of Nicaragua and El Salvador as well as Costa Rica and Guatemala have also documented the existence of a form of progressive CKD that, at that time, had no clear etiology^{369–372} (see also discussion in Chapter 76). The CKD that has been described has the following characteristics: (1) the location is relatively restricted to a specific geographic region, the coastal Pacific area of Central America; (2) there is no association with diabetes mellitus, hypertension, or other known CKD etiologies; (3) there is both male and younger age predominance; and (4) it occurs predominantly in low-income laborers working under extreme environmental conditions of heat and recurrent dehydration, such as agricultural workers performing cotton and sugarcane harvesting.³⁷³ The disease typically manifests as asymptomatic, progressive CKD, low-grade (<1 g/day) proteinuria, bland urine sediment, and mild hyperuricemia may be present. Kidney biopsies reveal tubulointerstitial disease and, in some cases, glomerular ischemia and glomerulosclerosis.³⁷⁴ Progression to ESRD occurs at a variable pace but because renal replacement therapy is scarce in the affected regions, mortality is very high.³⁷⁵

The etiology of Mesoamerican nephropathy is not known and is likely multifactorial. However, several potential risk factors have been proposed. These include heavy metals, agrochemicals, excessive use of NSAIDs, and consumption of contaminated alcohol. Underlying these mechanisms are repeated episodes of dehydration/volume depletion and perhaps episodes of rhabdomyolysis associated with strenuous labor.³⁷⁵ At this time, the best suggestion for prevention is to provide adequate hydration and limit exposure to extremes of heat. Further field studies investigating the pathogenesis of this devastating condition are ongoing.

A similar form of CKD associated with tubulointerstitial fibrosis has also been reported in Egypt, India, Tunisia, and Sri Lanka.³⁷⁶ Once again, the etiology of this condition is uncertain but epidemiologic data suggest that exposure to dietary and waterborne toxins may be causative.³⁷⁷ Specifically, rice and water levels of cadmium are elevated in affected regions.³⁷⁸ Further data are needed to understand the role of environmental toxins in the pathogenesis of this condition.

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BOARD REVIEW QUESTIONS

1. A 14-year-old male presents to your clinic with his third episode of gout in the past year. Prior to these episodes, he was healthy. He has been treated for past episodes with prednisone, nonsteroidal antiinflammatory drugs, and most recently was started on allopurinol 100 mg daily. His physical examination is notable for a normal blood pressure and except for a painful and swollen right ankle, his examination is otherwise normal. His family history is notable for several family members with gout and two family members are on dialysis. Laboratory work today reveals the following:

- Serum creatinine: 1.4 mg/dL (serum creatinine 1 year ago was 1.1 mg/dL)
- Uric acid: 9.7 mg/dL
- Urinalysis: negative for blood and protein
- Renal ultrasound: within normal limits

Mutations in which one of the following proteins are likely to be found in this patient?

- a. Mucin-1
- b. Renin
- c. Polycystin-1
- d. Uromodulin
- e. Voltage-gated chloride transporter

Answer: d

Rationale: The clinical presentation is most consistent with a form of autosomal dominant tubulointerstitial kidney disease (ADTKD), specifically due to mutations in the gene encoding uromodulin. Affected individuals present with early-onset gout and progressive chronic kidney disease. Urinalysis is typically normal, although low levels of proteinuria may be seen. When measured, the fractional excretion of uric acid is low in these patients. The diagnosis is confirmed with genetic testing. Although mutations in the renin protein can lead to ADTKD, this is a very rare condition and patients also display low-normal blood pressure, mild elevations of serum potassium, risk for developing volume depletion and prerenal azotemia, and hypoproliferative anemia with low erythropoietin levels that manifests early in life but resolve with adolescence. Mutations in mucin-1 can also lead to ADTKD; however, prominent gout is not seen. Mutations in polycystin-1 lead to autosomal dominant polycystic kidney disease and mutations in the voltage-gated chloride channel are associated with Dent disease characterized by tubular proteinuria, excess calcium in the urine, formation of calcium kidney stones, nephrocalcinosis, and chronic kidney failure.¹²⁶

2. You are following a 49-year-old female who has been on dialysis for 2 years after suffering rapidly progressive kidney disease due to weight-loss supplements. A renal biopsy at the time revealed severe interstitial fibrosis. Her dialysis course has been uneventful and she has no complaints. She is interested in transplantation and has been referred to her local center.

Which of the following would you recommend prior to transplantation?

- a. Standard pretransplant workup
- b. She should not be considered for transplant as the risk of disease recurrence is too high

c. Bilateral nephroureterectomy

- d. Yearly computed tomography studies of her kidneys
- e. Urine cytology performed every 6 months

Answer: c

Rationale: Aristolochic acid is one of the most potent carcinogens known to man and is associated with a very high risk of uroepithelial malignancies. The risk is so high that patients planning to undergo renal transplantation should be recommended to undergo bilateral nephroureterectomy once they begin dialysis therapy. Although screening cytology or imaging may be appropriate in lower-risk situations, the risk is so high in these patients that these tests may miss tumors and surgical cure is recommended.³¹⁰

3. A 57-year-old man presents to the renal clinic for evaluation of elevated serum creatinine (Cr) concentration. Serum creatinine concentration has increased from a baseline Cr of 1.0 mg/dL 1 year ago to 2.4 mg/dL. Chief complaints include dry mouth and dry eyes. He has no other medical problems. He sees his primary care physician annually and his screening is current. Serologic evaluation includes negative antinuclear antibodies, anti-double-stranded DNA, SS-A, and SS-B antibodies, but elevated immunoglobulin G class (IgG)4 subclass. Urinalysis shows 1+ protein and positive leucocyte esterase.

What renal histologic findings will most likely be seen in this case?

- a. Lymphoplasmacytic infiltrate and interstitial fibrosis in “storiform” pattern
- b. Mononuclear infiltrate with caseating granulomas
- c. Mononuclear infiltrate with associated glomerular lesion of minimal change
- d. Immunofluorescence with “antitubular basement membrane” antibodies
- e. Immunofluorescence with “full house” staining

Answer: a

Rationale: IgG4-related disorders are characterized by various organ system involvement, including the salivary glands, pancreas, retroperitoneum, and kidneys; elevated serum levels of IgG4 subclass antibodies; and tissue deposition of IgG4 in involved tissues. Middle-aged men are typically affected. The diagnosis of IgG4-related kidney disease is suspected when kidney dysfunction is observed in a patient with known or suspected IgG4-related systemic disease. The spectrum of kidney involvement includes acute or chronic tubulointerstitial nephritis, granulomatous interstitial nephritis, and obstructive renal disease due to retroperitoneal fibrosis (sometimes in the absence of hydronephrosis on imaging). Kidney histopathology is characterized by a lymphoplasmacytic interstitial infiltrate with a predominance of IgG4-positive plasma cells and interstitial fibrosis with a storiform pattern, which is typified by a cartwheel appearance of arranged fibroblasts and inflammatory cells. Treatment relies primarily on long-term glucocorticoid therapy; however, steroid-sparing agents such as mycophenolic acid or rituximab have been used with some success. Immunofluorescence with “antitubular basement membrane” antibodies is a rare finding in acute tubulointerstitial nephritis seen in the setting of methicillin, rifampin, allopurinol, and phenytoin. Immunofluorescence

with “full house” staining is seen in lupus nephritis. Mononuclear infiltrate with caseating granulomas is typical of tuberculosis. Mononuclear infiltrate with associated glomerular lesion of minimal change is typically seen with nonsteroidal antiinflammatory drugs–associated acute tubulointerstitial nephritis.²⁰²

4. A 71-year-old man with underlying stage 3A chronic kidney disease (estimated glomerular filtration rate = 47 mL/min per 1.73 m²) is hospitalized for upper arm cellulitis and is treated with intravenous piperacillin-tazobactam. After 9 days of therapy, serum creatinine concentration rises abruptly from baseline 1.8 mg/dL to 3.3 mg/dL. Despite intravenous fluids, kidney function continues to decline over the next 3 days. Complete blood cell count is within normal limits with white blood cells (WBCs) 9.0 and eosinophils 4%. Electrolytes are normal. Urinalysis reveals 1+ protein, 1+ blood, and 1+ leukocyte esterase. Urine sediment reveals 10–15 WBCs/high-power field (HPF), 3–5 red blood cells/HPF, and 1–2 WBC casts/low-power field. Urine eosinophils are >1% on Hansel stain and urine culture is negative. Renal ultrasound shows bilateral 15-cm kidneys with increased echogenicity and no hydronephrosis.

Which of the following will secure a diagnosis of acute tubulointerstitial nephritis as the cause of acute kidney injury in this patient?

- a. Leukocyturia with negative urine culture
- b. Positive urinary eosinophils
- c. WBC casts in the urine
- d. Kidney biopsy showing inflammatory interstitial infiltrate is required
- e. Large, echogenic kidneys on renal ultrasound

Answer: d

Rationale: Diagnosing acute tubulointerstitial nephritis as a cause of acute kidney injury is often very difficult. Many clinical, laboratory, and imaging tests suffer from inadequate sensitivity and specificity. Urine eosinophils are often ordered but seldom helpful as they suffer from false-positive and false-negative results. As a result, kidney biopsy is generally required to definitively diagnose this entity. Mixed inflammatory infiltrate (lymphocytes, plasma cells, eosinophils, macrophages) within the renal interstitium, tubulitis, and sometimes granuloma are diagnostic findings for acute tubulointerstitial nephritis. Glomeruli and blood vessels are spared, unless there is a concomitant renal process such as hypertension, diabetic nephropathy, or another renal disorder.¹⁷⁰